

Displace Him, Deploy Her? Couple's Asymmetric Labor Responses to Pension Eligibility

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October 6, 2026

Abstract

This paper challenges the consensus that when one spouse retires, the other is more likely to retire as well because of their complementarities in leisure. We show theoretically that when pension replacement rates are low and households depend heavily on home production, one spouses retirement can instead increase the other spouses labor supply, as they substitute toward market work to offset lost earnings rather than toward shared leisure. Using a regression discontinuity design in rural China, we find that when husbands become pension-eligible, their wives labor supply rises substantially. Eligibility pushes men out of higher-wage formal jobs into home production while their pension replaces only a small share of lost income. These results reconcile conflicting evidence across labor and development economics and highlight how pension design can unintentionally place a greater burden of old-age security on women.

1 Introduction

In many countries, generous pensions make retirement a choice. In settings with limited benefits and weak safety nets, eligibility can instead constrain continued employment. When eligibility forces exit from formal work without adequate replacement, older couples either accept a lower living standard or reallocate labor within the household. The question of how a couple jointly make retirement decisions is of particular relevance for pension policy design. As behavioral responses to retirement incentives often occur at the household level, ignoring such interdependence may leave significant welfare losses overlooked.

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Much of the existing evidence comes from high-income countries, where researchers have documented that spouses often synchronize their retirement decisions. In the United States, early studies found strong correlations in retirement timing between husbands and wives (Hurd, 1988; Blau, 1998), with nearly one-third of dual-earner couples leaving the labor force within a year of each other. Similar patterns of joint retirement have since been reported in Denmark (García-Miralles and Leganza, 2024), Norway (Kruse, 2021), Switzerland (Lalive and Parrotta, 2017), Australia (Atalay et al., 2019), the Netherlands (García and van Soest, 2022), and Germany (Moghadam et al., 2024). Another evidence from Norway also shows that retirement incentives can generate spillovers within couples: when one spouse becomes eligible for early retirement, the other spouse is more likely to exit the labor force at the same time through disability benefits (Johnsen et al., 2022).

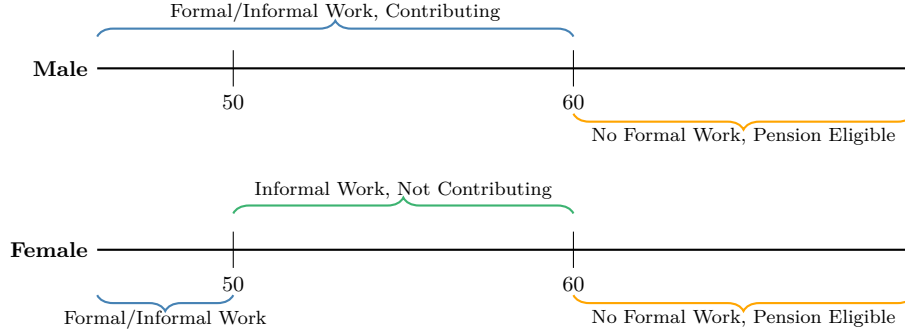
A common interpretation of joint retirement is based on the theory of *leisure complementarity*. In this view, couples derive greater utility from spending time together than from separate leisure, providing a natural explanation for their tendency to coordinate retirement. This framework, formalized in collective household models (Chiappori, 1992; Gustman and Steinmeier, 2000, 2009), has served as the foundation for a wide range of structural and reduced-form analyses (Michaud and Vermeulen, 2011; Atalay et al., 2019; Stancanelli and Soest, 2012; Lalive and Parrotta, 2017; Kruse, 2021; García-Miralles and Leganza, 2024; Michaud et al., 2020). Home production, as evidence suggests, is also a particularly pronounced channel for coordinated retirement behavior. This is reflected through the observed pattern of complementarity in non-market time. Time-use studies reveal asymmetric patterns: wives' retirement reduces husbands' home-production time, but not vice versa (Stancanelli and Soest, 2012). Moreover, retirement does not rebalance housework, since increases concentrate among women and single men rather than married men at their own retirement (Ciani, 2016). Additional studies report much smaller cross-effects of retirement or statistically insignificant spousal spillovers on wives' home production (Bonsang and Van Soest, 2020).

Taken together, the literature suggests that joint retirement cannot be explained solely by leisure complementarities. While some studies highlight coordinated leisure, others show that retirement often reallocates time toward home production, with limited evidence of more equal sharing of core household chores. Importantly, this evidence comes largely from high-income settings and may not generalize to poorer countries, where non-market time is more closely tied to home production and family care rather than leisure. This gap highlights the importance of examining how retirement behavior responds to institutions in lower-income settings with limited pensions.

Therefore, what will happen if a household member is displaced from the labor force by pension eligibility but endowed with inadequate pension replacement? We examine the effects of pension eligibility on household labor reallocation in rural China, where there is a unique design with staggered ages: men face *exit and pension eligibility at 60* while women face *exit at 50* but *eligibility at 60*. As shown in Figure 1, pension eligibility coincides with the statutory retirement age only for men, whereas women retire at 50 but cannot claim benefits until 60. The rules restrict access to

formal employment after the retirement age but do not bar informal work. This 10-year gap differs from urban China and from most other countries, giving us a built-in laboratory to separate *access-loss* effects from pure *transfer* effects, and to map each to testable predictions for intra-household labor reallocation.¹

Figure 1: Labor Market Transitions under the New Rural Pension Scheme



This figure illustrates the labor market transitions and pension eligibility timeline for rural residents under the NRPS. Both genders become pension-eligible at age 60, but face different formal employment constraints. Women typically exit formal employment at 50 but may continue informal work without pension contributions until age 60.

Our approach has two parts. First, we develop a theoretical framework that embeds age-based retirement constraints in a model with a CES aggregator over spouses non-market time. Following (Gronau, 1977; Chiappori, 1992), the framework captures how couples reallocate time between market work and home production when borrowing constraints and modest pensions restrict late-life consumption smoothing. We model pension eligibility as a discrete decline in formal earnings plus a small transfer, implying that spousal labor supply rises when leisure complementarities are weak and home production is an effective substitute.

Second, we implement an empirical strategy inspired by our model. We estimate using regression discontinuity design (RDD) and double regression discontinuity design (D-RDD) that exploits the age cutoffs for couples, allowing separate identification of own-eligibility effects and partner spillovers under standard RDD logic. Using CHARLS (2011-2018) (Zhao et al., 2014), we document a reversal of joint-retirement patterns: when husbands reach pension eligibility age, wives increase labor supply. This adjustment offsets the household income shortfall from husbands' mandated exit from formal jobs. At the threshold, wives' participation rises by 5.3-5.4 p.p., accompanied by a reallocation toward agricultural production. The response is larger in couples where husbands ever had a non-agricultural employed experience before 60 (our proxy for formal employment), underscoring the importance of earnings loss. Consistent with this mechanism, husbands' labor supply shift from non-agricultural to agricultural activities, highlighting home production as a key margin for smoothing consumption.

¹We observe a small number of rural pension enrollees contributing after the statutory retirement age. This reflects implementation timing: in some counties, individuals with fewer than 15 years until retirement were allowed to make up contributions to meet the 15-year contribution requirement, even though post-retirement contributions are typically not permitted.

Our study contributes to the emerging literature on retirement and household labor supply in low-income settings with weak social protection. While much of the existing evidence on joint retirement comes from OECD countries with generous pensions and flexible labor markets, our findings highlight how, in poorer contexts, pension eligibility can lead to labor reallocation within couples rather than coordinated exits. By documenting spousal asymmetries and the role of home production in rural China, we provide new evidence on how couples adapt to retirement shocks when formal insurance mechanisms are limited.

This pattern echoes the long-standing literature on the *Added Worker Effect*, a phenomenon first documented empirically during the Great Depression (Woytinsky, 1942; Humphrey, 1940) and later debated by Long (1958). The theoretical foundation is provided by Mincer (1962), showing that married women’s labor supply responds more strongly to transitory than permanent income changes. Building on this framework, Lundberg (1985) emphasized how liquidity-constrained households respond through spousal labor supply adjustments, while Layard and Nickell (1986) demonstrated how labor market institutions and rigidities shape these household responses. Modern evidence confirms substantial spousal responses across diverse contexts: Stephens Jr (2002) documents 3-5 percentage point increases in wives’ labor supply following husbands’ job displacement in the United States, Bredtmann et al. (2018) find effects ranging from 3 to 17 percentage points across European countries with stronger responses where welfare scheme is less generous, and Gong (2011) shows the effect is strongest among liquidity-constrained Australian households. In our rural Chinese context, this added-worker mechanism emerges directly from retirement policy design. When husbands face mandatory retirement at age 60 without adequate benefit replacement, wives expand agricultural and home business activities to stabilize household consumption.

The paper unfolds as follows. Section 2 sets the stage by examining China’s rural pension system, revealing a critical distinction: in this context, pension *eligibility* and conventional understanding of *retirement* are fundamentally different concepts, a divergence that drives our entire story. Section 3 develops our theoretical framework, showing how institutional context shapes household behavior and generates opposing predictions from standard models. Section 4 introduces our data from CHARLS and presents descriptive evidence of couples’ divergent retirement paths. Section 5 outlines our regression discontinuity strategy and identification assumptions. Section 6 reports the main results, showing wives increase labor supply when husbands reach eligibility while men show minimal response to wives’ eligibility. Section 7 dives deeper into the mechanisms at work, using heterogeneity analysis and time-use data to trace exactly how households reallocate labor across sectors and activities. Section 8 calculates welfare implications using structural estimates. Section 9 concludes with implications for pension reform in developing countries.

2 Institutional Background

The architecture of China’s social welfare system, including pensions, is fundamentally shaped by the hukou (household registration) system.² Pension enrollment is tied to hukou status, employment, and place of work, producing a dual system in which urban residents enjoy far more generous protection than their rural counterparts. According to official 2020 statistics, urban employee pensioners received average monthly benefits of about 2,900 yuan, while rural pensioners received only 170 yuan — a seventeen-fold gap.³ By comparison, monthly income differences between urban and rural residents were roughly 3,653 yuan versus 1,428 yuan, just a three-fold gap.⁴ This disparity implies replacement rates below 20% in rural areas, compared with about 80% in urban areas.⁵ As a result, inequality between urban and rural residents widens at retirement, and for many rural households, retirement from a formal job brings financial strain, making continued work or intra-household labor reallocation essential.

The New Rural Pension Scheme (NRPS) was introduced in 2009 to expand coverage to rural residents historically excluded from formal pension systems,⁶ and covering all counties across the country in 2012. Enrollment is voluntary for all rural hukou holders aged 16 or older who are not covered by the urban employee pension system. Participants choose fixed annual contribution levels between 100 and 500 yuan, but benefits are largely driven by a government-financed basic pension, initially set at 55 yuan per month in 2009 and raised to 88 yuan by 2018. Individuals already aged 60 at the programs inception received benefits immediately without any contributions.

The generosity of the NRPS is extremely limited. Benefits fall short of subsistence levels and are similar across most provinces, functioning instead as a minimal transfer. The design contrasts sharply with the contributory, earnings-linked urban employee pension.

Hukou-based pension rules in China separate retirement from pension eligibility for rural women but not for rural men or the urban population, illustrated by Figure 1. Most rural women must retire from formal employment at age 50 yet cannot claim pension benefits until age 60, creating a ten-year coverage gap. By contrast, rural men retire and become pension eligible at the same age at 60, mirroring the urban employee system. For men, this simultaneity produces a double shock at age 60, the loss of access to formal employment alongside the onset of a minimal pension.⁷

These institutional features imply that rural residents can extend working lives only by shifting into informal sectors after formal retirement. Some evidence (Giles et al., 2011) suggests that

²The hukou system categorizes citizens as rural (agricultural) or urban (non-agricultural), historically determining access to pensions, healthcare, education, and employment. Although reforms have relaxed some restrictions, hukou remains central to rural-urban inequality in welfare entitlements (Wang, 2005; Cheng and Selden, 1994; Wu and Treiman, 2004; Chan and Zhang, 1999; Chan, 2009).

³See https://www.stats.gov.cn/sj/zxfb/202509/t20250928_1961404.html and <https://www.nhc.gov.cn/11jks/c100157/202110/b365898ae2a4406d91e1b1ecd10ca911.shtml>.

⁴See https://www.stats.gov.cn/xxgk/sjfb/zxfb2020/202101/t20210118_1812464.html

⁵For comparison, OECD replacement rates are about 70-80% (Chen et al., 2021; Wang, 2006).

⁶Before 2009, fragmented local pension systems existed with similar age requirements but relied primarily on personal contributions with limited local government support. These were automatically integrated into the NRPS upon its introduction.

⁷In some localities, it is technically possible to keep contributing after the retirement age, but benefits are delayed.

rural elderly work significantly longer than their urban counterparts, frequently remaining active in agriculture well into their 70s. Other studies emphasize household-level adjustments: some literature (Lei et al., 2012) suggests that intergenerational transfers partially compensate for weak pensions, while further evidence (Ning et al., 2016) indicates that receiving NRPS pensions does not reduce, and may even increase, elderly labor supply. Taken together, these findings underscore how weak benefits and rigid rules can sustain elderly labor force participation and possibly drive intra-household reallocation, in contrast to OECD settings where pension eligibility typically facilitates labor force exit.

3 Conceptual Framework

We model a rural Chinese household’s labor allocation when facing husband’s mandatory retirement at age 60.⁸ The household allocates time across formal work, informal work, and non-market activities (which includes leisure, care work or home production). Individuals are subject to age-based employment constraints and pension benefits.

Building on the household production and time allocation framework in Becker (1965); Gronau (1973, 1977) and the collective household perspective from Chiappori (1992); Rogerson and Wallenius (2019); Michaud et al. (2020), we incorporate population heterogeneity in formal job access. We provide a unitary household model with pooled resources with comparative statics that match the collective benchmark under standard pooling.

Household Structure A household (husband h , wife w) allocates each spouse’s unit time among formal F_i , informal (incl. agriculture) I_i , and non-market S_i :

$$F_i + I_i + S_i = 1, \quad i \in \{h, w\}.$$

Informal opportunities always pay $r > 0$. Formal job provides a higher return $\bar{w} > r$ before eligibility, and formal workers lose that access after pension eligibility age. Let $s \in \{\text{pre}, \text{post}\}$ denote the state relative to eligibility. A transfer of $B > 0$ arrives post.

Household Types. Only a fraction $p \in (0, 1)$ of rural households have obtained formal employment experience. In *pre*, both market activities (formal, informal) are potentially accessible to the husband in a subset of households (Type H); in *post*, the formal activity is *not* accessible to him (loss of access). In Type L households, the husband never obtained formal job opportunity in either state. In our sample, most spouses are within five years of each others age, so by the time husbands reach 60, their wives have already passed their own retirement age. Therefore wives operate only in informal employment around the relevant states and their market return is a constant $r > 0$.⁹

⁸Our model focuses on the husband’s mandatory retirement as the household’s main earner, but we also discuss potential effects at the wife’s retirement age in the Results section.

⁹This $\{L, H\}$ type heterogeneity captures the data documented in Section 2: not all rural men work in formal employment before 60, and those who do experience a sharp income loss at retirement. The parameter p entails both factors from the labor market structure (availability of formal jobs) and may also reflect individual characteristics (education, location, and skills).

- **Type H (access loss):** $w(\text{pre}) = \bar{w}$, $w(\text{post}) = r$
- **Type L (no access change):** $w(\text{pre}) = w(\text{post}) = r$.

We use $w(s)$ to denote the husband's accessible market return in state $s \in \{\text{pre}, \text{post}\}$; the wife's accessible return is $r > 0$ throughout.

Preferences and aggregator. Preferences are modeled on a unitary household level, $U(C, S_h, S_w) = \alpha \log C + \phi \log G(S_h, S_w)$, where

$$G(S_h, S_w) = [\gamma S_h^\rho + (1 - \gamma) S_w^\rho]^{1/\rho},$$

with elasticity $\sigma \equiv 1/(1 - \rho) > 0$ and weights $\alpha, \phi > 0$, $\gamma \in (0, 1)$. Under linear returns and no frictions, each spouse specializes in the accessible activity with highest return.

Two-stage budgeting and the CES unit cost. Under linear returns and specialization, we can apply the standard full-income transformation: each spouse inelastically supplies one unit of time to market at their best accessible return (husband at $w(s)$, wife at r), and then buys back non-market time at prices $(w(s), r)$.

This delivers *full income* $Y(s) = w(s) + r + B(s)$ and a unit cost $\Pi(w(s), r)$ for the CES composite (G). Cost-minimizing production of one unit of G under prices (w, r) yields the CES unit cost

$$\Pi(w(s), r) = [\gamma^\sigma w(s)^{1-\sigma} + (1 - \gamma)^\sigma r^{1-\sigma}]^{\frac{1}{1-\sigma}},$$

so the household chooses (C, G) from

$$\max_{C, G} \alpha \log C + \phi \log G \quad \text{s.t.} \quad C + \Pi(w(s), r) G \leq Y(s).$$

Cobb-Douglas in (C, G) implies

$$C^*(s) = \frac{\alpha}{\alpha + \phi} Y(s), \quad G^*(s) = \frac{\phi}{\alpha + \phi} \frac{Y(s)}{\Pi(w(s), r)}. \quad (1)$$

By Shephard's lemma (cost shares inside G),

$$S_h^*(s) = \frac{\phi}{\alpha + \phi} Y(s) \frac{\gamma^\sigma}{\Pi(w(s), r)^{1-\sigma} w(s)^\sigma}, \quad S_w^*(s) = \frac{\phi}{\alpha + \phi} Y(s) \frac{(1 - \gamma)^\sigma}{\Pi(w(s), r)^{1-\sigma} r^\sigma}, \quad (2)$$

and $L_i^*(s) = 1 - S_i^*(s)$. Indirect utility is

$$V(w(s), r, B(s)) = (\alpha + \phi) \log Y(s) - \phi \log \Pi(w(s), r) + K. \quad (3)$$

A compact derivation of Π and (2) appears in Appendix A (Deriving the CES unit cost).

3.1 Main Theoretical Predictions

Define $g \equiv \bar{w}/r > 1$ and $R \equiv B/\bar{w} \in [0, 1]$. *Formal-informal wage ratio.* $g \equiv \bar{w}/r$ measures the formal wage premium. Higher g means larger earnings shortfall when formal access is removed; *Replacement rate* $R \equiv B/\bar{w}$ measures how well the pension compensates for lost formal earnings.

Proposition 1 (Added-Worker Threshold in Type H Households). *For Type H households with $w(\text{pre}) = \bar{w} > r$ and $w(\text{post}) = r$, and $U(C, S_h, S_w) = \alpha \log C + \phi \log G(S_h, S_w)$ with $G(S_h, S_w) = [\gamma S_h^\rho + (1-\gamma)S_w^\rho]^{1/\rho}$ and $\sigma \equiv 1/(1-\rho) > 0$. The wife's market time adjustment $\Delta L_w \equiv L_w^*(\text{post}) - L_w^*(\text{pre})$ satisfies*

$$\Delta L_w > 0 \iff R < R^*(\sigma; g, \gamma),$$

where

$$R^*(\sigma; g, \gamma) = \frac{g+1}{g} \cdot \frac{\gamma^\sigma + (1-\gamma)^\sigma}{\gamma^\sigma g^{1-\sigma} + (1-\gamma)^\sigma} - \frac{2}{g}.$$

Moreover, the threshold exhibits the following comparative statics:

- (i) R^* is strictly increasing in g for all $\sigma \geq 1$ and $\gamma \in (0, 1)$;
- (ii) R^* is increasing in σ when $\gamma = \frac{1}{2}$ (and in a neighborhood thereof).

When the husband loses access to high-paying formal employment ($\bar{w}$) at age 60, the household faces two competing forces: (a) the income decline (smaller Y) pushes the wife to increase market work; (b) the shadow price of the non-market composite G , given by $\Pi(w, r)$, falls as w drops from $\bar{w}$ to r , increasing the relative attractiveness of home time. The pension B provides partial insurance. The threshold $R^*(\sigma; g, \gamma)$ is the replacement rate at which these two forces cancel out. When $R < R^*$, the income effect dominates and the wife *increases* labor supply; when $R > R^*$, the price effect dominates and she *reduces* work. The threshold rises with g , implying that a larger income gap would require a higher replacement rate to suppress the added-worker response. When σ is high (more substitutable non-market time, more flexible division of labor), the wife can more easily “fill in,” expanding the added-worker region where $R < R^*$.¹⁰

Corollary 1 (Sectoral Reallocation). *Under Type H, in post state the husband supplies zero formal hours and any market time is informal/agricultural. Wife's added labor also appears in informal/agricultural activities where access constraints do not bind.*

Proposition 2 (Pure Transfer in Type L Household). *If accessible market returns are unchanged across states (in particular, $w(\text{pre}) = w(\text{post})$ and wife's accessible return remains r), then eligibility operates as pure transfer:*

$$C^*(\text{post}) - C^*(\text{pre}) = \frac{\alpha}{\alpha + \phi} B > 0, \quad L_i^*(\text{post}) - L_i^*(\text{pre}) < 0 \quad (i \in \{h, w\}),$$

and the relative split of (S_h, S_w) is unchanged.

¹⁰Detailed derivations can be found in Appendix A.

When formal access is removed (Type H), the shadow price of G falls via Π , but the household loses formal earnings. If pension B is too small relative to exposure g and substitutability σ , the income shortfall dominates and the wife increases market work (Prop. 1). When access is unchanged (Type L, or wife’s own eligibility), eligibility simply relaxes the budget constraint and both spouses reduce market work proportionally (Prop. 2). However, a primary test of this pure transfer effect is through household consumption, which should unequivocally increase.

Heterogeneity in ϕ and response magnitude Changing ϕ does not move the decision boundary $R^*(\sigma; g, \gamma)$ nor the sign predictions in Proposition 1; the magnitude of the household labor allocation response is scaled by ϕ . The income/transfer channel becomes stronger as ϕ rises, independently of σ ; the price (relative-wage) channel has a larger *absolute* impact when ϕ is larger, except in the Cobb-Douglas case $\sigma = 1$ where this channel vanishes. Since $L_w = 1 - S_w$, the magnitude of ΔL_w scales accordingly. Finally, the marginal propensity to consume out of the transfer is $\text{MPC}_C = \alpha/(\alpha + \phi)$, which decreases in ϕ . More analysis and a proof can be found in the appendix A.8.

3.2 Visualization

To visualize heterogeneity in formal-sector exposure among Type H households, suppose their formal wage premia $g = \bar{w}/r$ lie in an interval $[g_L, g_H]$. We treat g_L and g_H as lower and upper bounds of this exposure range.

We use decision maps to summarize the model’s prediction for the sign of the wife’s labor response at the pension cutoff. Figure 2 plots the decision boundaries in the (σ, R) space for Type H households. The two curves represent the threshold $R^*(\sigma; g)$ at the lower and upper bounds of the formal wage premium range $[g_L, g_H]$. These envelopes partition the parameter space into three regions: below the lower curve (green), the income effect dominates and wives increase labor for all exposure levels; above the upper curve (red), the price effect dominates and wives reduce labor regardless of g ; in the intermediate zone (orange), the response depends on the household’s specific formal wage premium. The vertical dashed line marks the Cobb-Douglas case ($\sigma = 1$), where the price channel vanishes and only the income channel operates.

Type H households: Wife's labor response regions

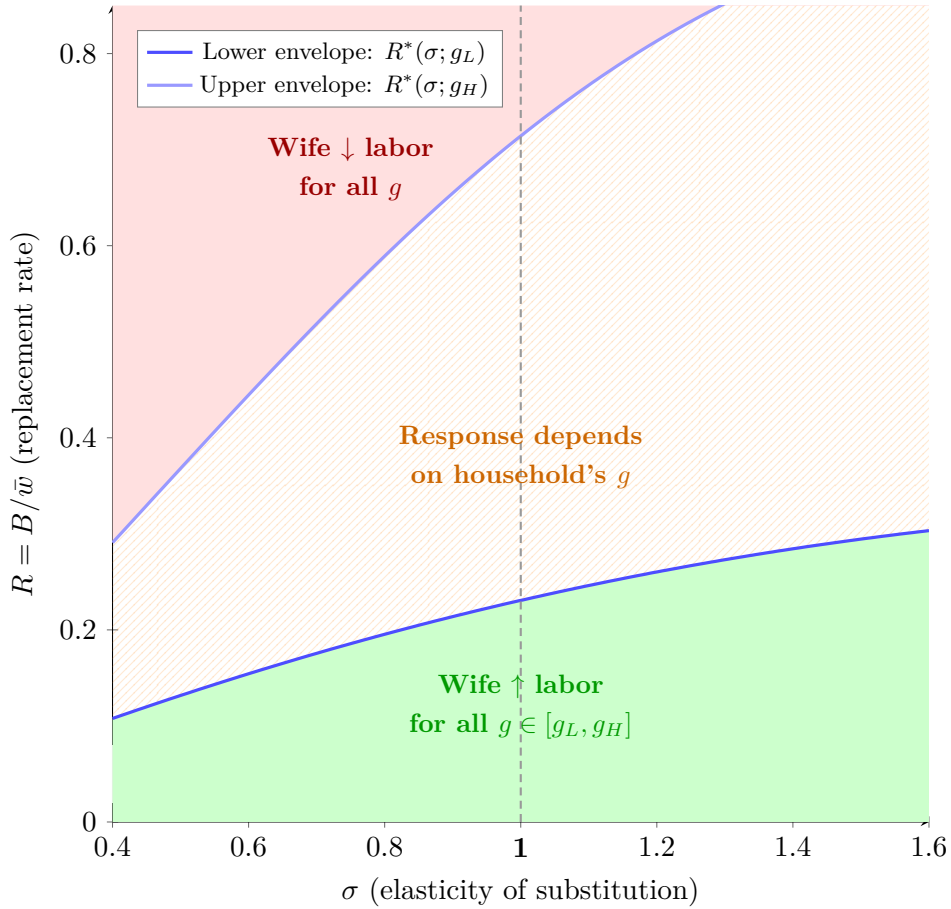


Figure 2: Type H households (husband loses formal access at eligibility). The two curves are decision-boundary envelopes $R^*(\sigma; g_L)$ and $R^*(\sigma; g_H)$ from Proposition 1.

Figure 3 illustrates how the threshold replacement rate $R^*(g)$ varies with the formal wage premium g at two levels of substitutability ($\sigma = 1.2$ vs. $\sigma = 0.8$). Households facing larger income shocks require higher pension replacement to avoid triggering the added-worker response. When the husband's formal wage premium is high (g large), the loss from mandatory retirement is substantial, raising the threshold R^* , below which the income effect dominates.

When spouses can flexibly reallocate household tasks, the added-worker region expands because wives can more easily step in to offset the husband's earnings loss. This comparative static directly maps to our empirical strategy in Section 6, where we split the sample by husbands' pre-retirement employment sector to proxy for variation in g .

To provide a more complete picture, Figure 4 shows labor response in both types of households (L and H). The left panel (Type H: access loss) reproduces the map in Figure 2. The right panel (Type L: transfer-only) is uniformly red: with no access change, eligibility acts as a pure income effect and the wife's labor falls for any $R > 0$.

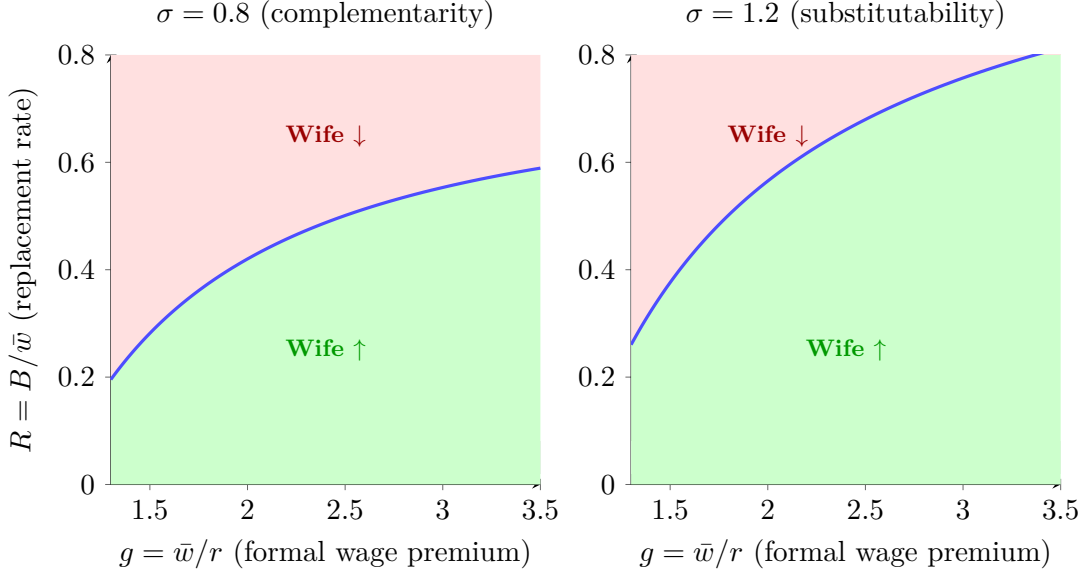


Figure 3: Decision boundaries $R^*(g)$ at fixed σ .

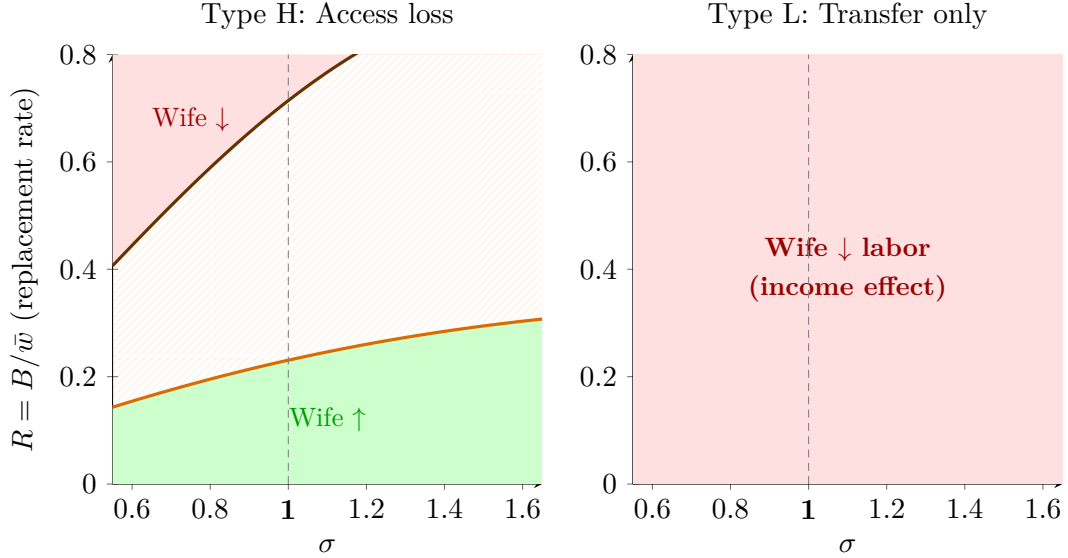


Figure 4: Decision Map for Household type L and H.

These visualizations together illustrate the core trade-off in Type H households: the income loss from leaving formal employment (stronger when g is high) competes with the reduced shadow price of non-market time. When replacement rates are low (R small), the income shortfall dominates, inducing wives to increase labor supply (green regions). As R rises, the transfer smooths consumption and weakens the insurance motive, eventually allowing both spouses to reduce market work (red regions). The heterogeneity captured by these decision maps with stronger spousal responses predicted among high- g couples guides our empirical strategy in Section 6.

Two-period extension Appendix B develops a minimal two-period extension with savings/borrowing. It delivers a sufficient statistic condition for the wife’s response, nesting our static threshold and rationalizing larger spousal responses when liquidity is tight.

4 Data

Our analysis draws on the China Health and Retirement Longitudinal Study (CHARLS), a nationally representative biennial survey of Chinese residents aged 45 and older. CHARLS is part of the international family of Health and Retirement Studies, designed to be comparable with the HRS in the U.S. and similar surveys worldwide (Zhao et al., 2014).

The baseline national survey, conducted between 2011 and 2012, interviewed roughly 17,700 individuals from about 10,200 households, sampled from 150 counties and districts across 28 provinces. The survey employed a multistage, stratified probability-proportional-to-size (PPS) sampling design: counties/districts were chosen in the first stage, villages or urban communities in the second, and within each household, one age-eligible respondent (45+) was randomly selected, with spouses automatically included regardless of age. Follow-up interviews were carried out in 2013, 2015, and 2018, with response rates consistently above 80%.

CHARLS collects rich information on demographic characteristics, family structure, employment, pension participation, health, income, consumption, and time use. For this paper, we primarily use the detailed modules on labor supply in agricultural and non-agricultural work, pension eligibility and receipt, and time spent, including caregiving and household activities. Our analysis uses the 2011, 2013, 2015, and 2018 survey waves.

Sample Construction

We focus on couples both enrolled in the New Rural Pension Scheme (NRPS) or its successor, the Urban-Rural Pension Scheme.¹¹ To ensure comparability, we restrict the sample to married or cohabiting individuals aged 45 to 75 whose spouses also fall within this age range, and exclude anyone simultaneously enrolled in the urban employee pension system or never worked for at least three months during their lifetime.

This focus on rural pension enrollees is motivated by three considerations. First, the rural pension system provides substantially lower replacement rates (around 20%) than urban schemes, making it particularly informative for analyzing labor supply responses when benefits alone cannot sustain retirement (Chen et al., 2021). Second, the uniform eligibility age of 60 for both genders provides cleaner identification than urban pensions, which impose gender- and occupation-specific retirement ages. Third, rural residents typically remain in the labor force longer than their urban counterparts, with participation rates exceeding 70% for men aged 60 to 65, allowing adjustments at pension eligibility to be more readily observed (Giles et al., 2011).

Following the retirement spillovers literature (Lalive and Parrotta, 2017; García-Miralles and Leganza, 2024), we restrict attention to individuals aged 45-75 to capture behavior in the 15 years

¹¹We also include individuals enrolled in the Old Rural Pension Scheme in 2011 and 2013, as they faced identical retirement and pension eligibility ages and were automatically converted to the NRPS.

before and after eligibility. We pool observations across survey waves and treat the data as repeated cross-sections for our regression discontinuity design (Battistin et al., 2009; García-Miralles and Leganza, 2024).

Variable Construction

Labor Force Participation: We classify an individual as participating in the labor force if they report engaging in agricultural work, wage employment, self-employment, or unpaid family business work during the past year. This definition adopts a low threshold to capture the intermittent and seasonal nature of rural work, while excluding purely nominal participation.

Sectoral Employment: We distinguish between agricultural work (farming on own land or as hired labor) and non-agricultural work (wage employment or self-employment in non-farm activities). This distinction is central, as age-based retirement rules and formal employment barriers primarily constrain non-agricultural work.

Pension Receipt: Pension receipt is identified through self-reported rural pension benefit take-up. As shown in Figure 5, coverage rises sharply from near zero to roughly 50% at age 60, providing strong first-stage variation for our identification strategy.

Time Allocation: For mechanism analysis, we use detailed data on weekly hours devoted to: (i) agricultural work on own farm,¹² (ii) non-agricultural work, (iii) grandchild care, (iv) elderly parent care, and (v) leisure activities.¹³ To accommodate zeros in time (and later in income), we estimate effects in proportional terms using the inverse hyperbolic sine (IHS) transformation (Burbidge et al., 1988).

Income Measures: We construct monthly income from employment, as well as household-level measures of food consumption, agricultural production, and home business activities.

Control Variables: We control for both spouses years of education. All specifications include province-by-year fixed effects to capture regional heterogeneity and policy timing.

To probe the participation mechanism, we examine heterogeneity. Propositions 1 and 2 predict stronger spousal insurance where retirement entails a larger earnings loss. We classify husbands by whether they report any non-agricultural employment experience before age 60 and split the sample accordingly. This partition proxies exposure to formal employment in rural China context¹⁴ and mirrors exposure-based subgrouping by former sector in displacement settings (Jacobson et al., 1993).

Acknowledging Data Limitations

Several limitations merit discussion. First, while CHARLS maintains high response rates, attrition may be non-random with respect to health and mobility. Second, survey quality issues remain: pension enrollment, income, non-agricultural employment experience and time-use questions have relatively low response rates; the 2015 wave omits time spent on eldercare; in some cases, husbands report answers while wives do not. We address these challenges by inferring missing responses

¹²We exclude work on others farms, as it was not collected in the 2015 and 2018 waves.

¹³Because daily leisure time was reported in ranges (e.g., < 2 hours or ≥ 2 hours), we use the reported number of days per week of light-intensity physical activity instead.

¹⁴*Balancing Workers Protection and Labor Market Flexibility in China*. World Bank Group, 2021

from other questions related to labor force participation and by focusing on couples who are both enrolled in the program. Third, employment and income measures rely on recall over the past year, potentially introducing measurement error that would attenuate our estimates.

4.1 Descriptive Statistics

Table 1: Descriptive statistics by gender

Spouse's Age	45–59	60–75	Difference	RDD Coefficient	Std. Error
Panel A: Male					
Education	6.82	5.12	1.70	0.13	0.15
Spouse's Education	4.31	2.14	2.17	0.05	0.14
Self report health	3.20	3.37	-0.17	0.01	0.05
Number of parents	1.01	0.24	0.77	0.04	0.04
Number of grandchildren	2.49	3.05	-0.55	0.02	0.10
Support from children	1,994.43	2,810.27	-815.83	-444.46	308.70
Observations	6,803	4,428		11,231	
Panel B: Female					
Education	4.57	2.25	2.32	0.15	0.16
Spouse's Education	7.17	5.10	2.07	0.17	0.18
Self report health	3.35	3.50	-0.15	-0.04	0.05
Number of parents	1.09	0.27	0.81	0.03	0.04
Number of grandchildren	2.35	3.07	-0.72	-0.04	0.10
Support from children	1,909.72	1,778.71	131.01	-137.21	266.53
Observations	5,874	5,413		11,287	

Self-report health ranges 1 (very good) to 5 (very bad). Support from children is the financial support received from all children per year. Mean values and t-test differences are reported. Coefficient and standard error are referred to the spouse's pension eligibility age (dummy variable) of the quadratic Double RDD. Standard errors clustered at the individual level. Standard errors in last column. * $p < .1$, ** $p < .05$, *** $p < .01$.

Table 1 reports descriptive statistics for individuals by spouses age relative to pension eligibility, with men shown in the upper panel and women in the lower panel. Columns 1-3 present mean values and t-tests for differences between those whose spouse is younger than 60 and those whose spouse is 60 or older. Columns 4 and 5 provide regression discontinuity (RDD) estimates testing whether observed characteristics are balanced at the eligibility threshold.

Several patterns emerge. First, education levels decline sharply with age, reflecting Chinas rapid educational expansion. For example, men with spouses aged 45-59 average 6.82 years of schooling compared with 5.12 for those with spouses aged 60-75; the gradient is even steeper for women (4.57 vs. 2.25 years). Second, and crucial for identification, predetermined characteristics show no evidence of discontinuity at age 60. The balance tests in the 4th column confirm that education, health, and family structure are continuous across the threshold for both men and women. This validates the RDD continuity assumption necessary for causal inference (Hahn and Kuersteiner,

2002; Lee and Lemieux, 2010).

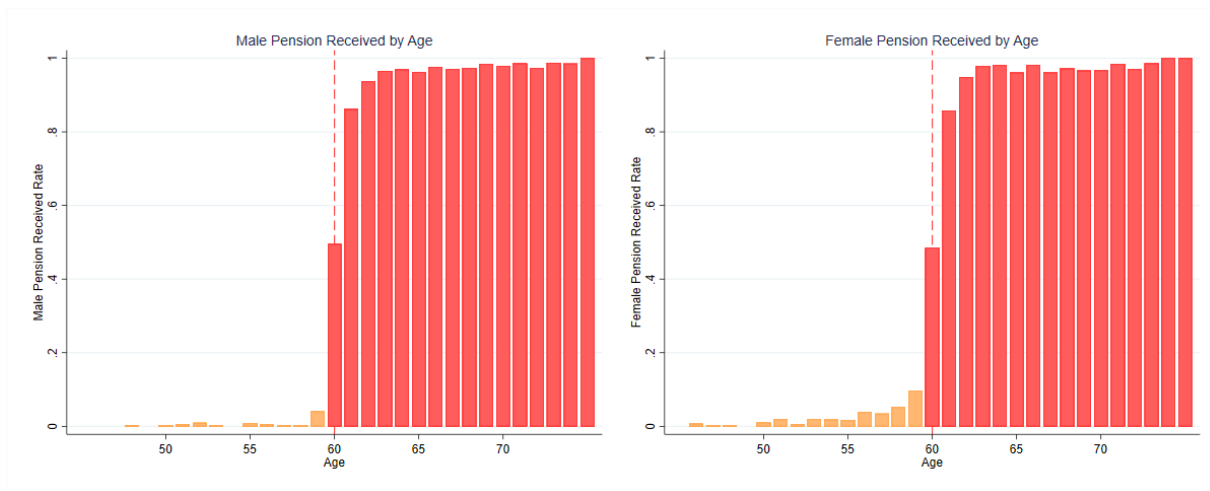


Figure 5: Percentage of Rural Pension Scheme Enrollees Receiving Pension

Figure 5 further illustrates the first stage: pension receipt rises sharply at age 60, jumping from near zero to about 50% for both genders. The uniform eligibility rule explains the nearly identical discontinuity for men and women. Although immediate take-up is incomplete due to administrative delays and limited awareness, the discontinuity is large enough to generate strong identification.

4.2 Descriptive Evidence

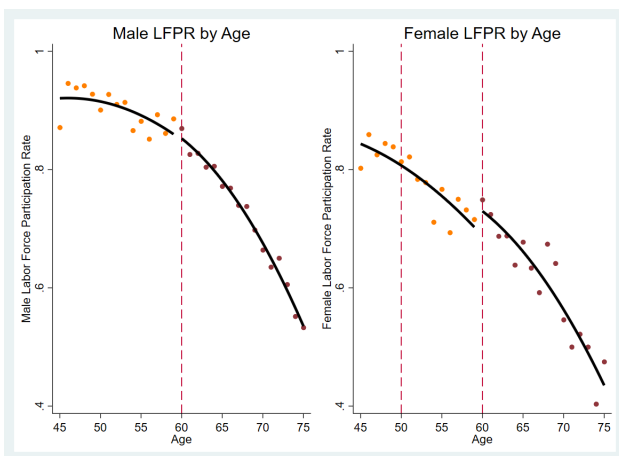


Figure 6: Labor Force Participation Rate by Age



Figure 7: Labor Force Participation Rate by Spouse's Age

Figures 6 and 7 provide visual evidence for our central hypothesis. Figure 6 shows that labor force participation (LFP) declines gradually with an individual's own age, with no visible discontinuity at the age-60 pension eligibility threshold. This indicates that own eligibility has little immediate effect on labor force exit. By contrast, Figure 7 displays clear and asymmetric discontinuities at the

spouses eligibility age: wives LFP rises sharply when husbands turn 60, while husbands LFP shows only a modest decline when wives become eligible. These patterns highlight husbands pension eligibility as the primary trigger for household labor supply adjustments in rural China, contrary to the joint retirement patterns emphasized in the OECD literature, but consistent with evidence on strong spousal spillovers from men to women.

5 Empirical Strategies

Following prior work (Battistin et al., 2009; García-Miralles and Leganza, 2024), we pool individuals from CHARLS for the period 2011-2018. This approach is well suited to regression discontinuity designs (RDD), which depend on the validity of the assignment rule rather than the panel structure of the data. In our context, pension eligibility is determined solely by age, a predetermined characteristic that individuals cannot manipulate, ensuring the credibility of the design.

We implement a regression discontinuity (RDD) design to estimate the causal effect of an individuals own or their spouses pension eligibility on labor supply outcomes. The framework builds on the potential outcomes approach (Rubin, 1974; Imbens and Lemieux, 2008; Seibold, 2021) and is adapted to capture spousal spillovers, allowing us to identify how one partners eligibility status influences the others labor supply decisions.

Let $Y_i(1)$ and $Y_i(0)$ denote the potential labor supply outcomes for individual i if their spouse is eligible or ineligible for a pension, respectively. We define the treatment variable D_i as:

$$D_i = \begin{cases} 1 & \text{if } A_i \geq \bar{a} \\ 0 & \text{if } A_i < \bar{a} \end{cases} \quad (4)$$

where A_i is the individual's age and $\bar{a}$ is the pension eligibility threshold (60 years in our context for male). The observed outcome is expressed as:

$$Y_i = Y_i(0) + [Y_i(1) - Y_i(0)]D_i \quad (5)$$

Our parameter of interest is the average treatment effect at the threshold:

$$\tau = E[Y_i(1) - Y_i(0)|A_i = \bar{a}]. \quad (6)$$

To estimate this effect, we assume that the conditional expectation of the potential outcomes is continuous in A_i at $\bar{a}$:

$$E[Y_i(0)|A_i = a] = f_0(a), \quad (7)$$

$$E[Y_i(1)|A_i = a] = f_1(a), \quad (8)$$

where $f_0(\cdot)$ and $f_1(\cdot)$ are continuous functions. Under these assumptions, τ can be identified as:

$$\tau = \lim_{a \downarrow R} E[Y_i | A_i = a] - \lim_{a \uparrow \bar{a}} E[Y_i | A_i = a]. \quad (9)$$

To account for heterogeneity, we specify the following model:

$$Y_i = \alpha + \tau D_i + f(A_i - \bar{a}) + \beta X_i + \delta(D_i \times X_i) + \varepsilon_i, \quad (10)$$

where:

$$f(A_i - \bar{a}) = \sum_{k=1}^K \theta_k (A_i - \bar{a})^k + \sum_{k=1}^K \phi_k D_i (A_i - \bar{a})^k. \quad (11)$$

Here, X_i is a vector of covariates, β captures their direct effects, and δ allows for heterogeneous treatment effects. The function $f(\cdot)$ is a flexible polynomial that controls for the running variable on either side of the threshold.

We estimate Equation (10) using local linear regression within a bandwidth h around the threshold taking into account both the bias and the variance (Imbens and Kalyanaraman, 2012; Imbens and Wager, 2019):

$$\hat{\tau} = \arg \min_{\tau, \alpha, \beta} \sum_{i: |A_i - \bar{a}| \leq h} (Y_i - \alpha - \tau D_i - \beta(A_i - \bar{a}) - \gamma D_i (A_i - \bar{a}))^2 K\left(\frac{A_i - \bar{a}}{h}\right), \quad (12)$$

where $K(\cdot)$ is a kernel function.

Unlike previous studies that focus on the impact of one's own retirement, we also implement a regression discontinuity design around the spouse's pension eligibility age:

$$Y_i = \alpha + \gamma \times 1(A_j \geq \bar{a}_j) + f_j(A_j) + f_j(A_j) \times 1(A_j \geq \bar{a}_j) + \eta_{ij}, \quad (13)$$

where A_j is the age of individual i 's spouse, $\bar{a}_j$ is the spouse's pension eligibility age, and $A_j \geq \bar{a}_j$ represents the spouse's pension eligibility indicator. The function $f_j(\cdot)$ describes how labor force participation evolves with the spouse's age relative to their pension eligibility age.

Based on our theoretical model, a spouse's pension eligibility may influence the optimal timing for exiting the labor market given the current rural pension system in China. Some studies argue that rural pensions in China may not encourage elderly individuals to withdraw from the labor market (Ning et al., 2016). This leaves room to explore the spillover effects of the spouse's pension eligibility that could affect labor supply decisions.

Therefore, we also estimate the effect using a double regression discontinuity (D-RDD) design (Stancanelli and Soest, 2012; Lalive and Parrotta, 2017) and allow interactions between the running

variable and the treatment indicator:

$$\begin{aligned}
Y_i = & \alpha + \gamma_1 \mathbb{1}(A_i \geq \bar{a}) + \gamma_2 \mathbb{1}(A_j \geq \bar{a}) \\
& + f_1(A_i - \bar{a}) + f_2(A_j - \bar{a}) + f_1(A_i - \bar{a}) \times \mathbb{1}(A_i \geq \bar{a}) \\
& + f_2(A_j - \bar{a}) \times \mathbb{1}(A_j \geq \bar{a}) + \beta X_i + \varepsilon_{ij}.
\end{aligned} \tag{14}$$

This model includes an RDD specification for both partners compared to the previous model, where γ_1 and γ_2 represent the effect of own and spousal pension eligibility.

We estimate this model using local linear regression within a bandwidth h around the threshold:

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$$\hat{\tau} = \arg \min_{\tau, \alpha, \beta} \sum_{i: |A_i - \bar{a}| \leq h, |A_j - \bar{a}| \leq h} (Y_i - \alpha - \tau_1 D_i - \tau_2 D_j - \beta_1(A_i - \bar{a}) - \beta_2(A_j - \bar{a}))^2 K\left(\frac{A_i - \bar{a}}{h}, \frac{A_j - \bar{a}}{h}\right),$$

where $K(\cdot)$ is a bivariate kernel function.

The D-RDD serves as a robustness check to the main results using simple RDD specification and analyze the complex dynamics of household retirement decisions.

6 Results

6.1 Main Finding: Asymmetric Response

Our theoretical framework in Section 3 makes predictions in Proposition 1 and Corollary 1 that when pension eligibility coincides with mandatory exit from formal employment, households will reallocate labor to insure against the income shock.

We start by documenting the main finding. Table 2 reports results from the traditional single RDD specification that conditions on own or spousal pension eligibility.

The results are clear and reveal an explicit gender asymmetry. As shown in Panel A, when a husband reaches pension eligibility, his wife's labor force participation increases by 5.3 p.p. ($p < 0.05$). In contrast, wife's own pension eligibility has small and statistically insignificant effect on her labor supply. This near-zero response is consistent with the pure transfer prediction of Proposition 2, as her access to formal labor markets is already constrained. This asymmetry overall points to a spousal insurance mechanism as predicted in Proposition 1.

For men in Panel B, the effects of either their own or their wife's eligibility are small and not statistically significant in our specifications. These findings are robust to various model specifications, as confirmed by the Double RDD estimates in Table 11 and Table 12.¹⁶

¹⁵In this double-RDD setting, bandwidth choice is non-standard because imposing a bandwidth for one spouse simultaneously selects the partner. For comparability across specifications, we adopt a fixed bandwidth around each cutoff and apply a triangular kernel.

¹⁶We exclude same-age couples in Table 12 to provide a robustness check because own and spousal eligibility coincide most frequently when the age gap is zero, which can possibly induce mutual contamination and bias estimates.

Table 2: Single regression discontinuity design on couples' labor force participation

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Women's Labor Force Participation						
Spouse's pension eligibility	0.054** (0.02)	0.053** (0.02)	0.053** (0.02)			
Own pension eligibility				0.027 (0.02)	0.025 (0.02)	0.018 (0.02)
Controls	No	Yes	Yes	No	Yes	Yes
Year $\times$ Province	No	No	Yes	No	No	Yes
Observations	11,049	10,958	10,956	11,004	10,921	10,919
Ineligible Group Mean	0.787	0.787	0.787	0.781	0.781	0.781
Adj R-squared	0.015	0.014	0.045	0.016	0.016	0.049
Panel B: Men's Labor Force Participation						
Spouse's pension eligibility	-0.019 (0.02)	-0.020 (0.02)	-0.021 (0.02)			
Own pension eligibility				-0.014 (0.02)	-0.012 (0.02)	-0.011 (0.02)
Controls	No	Yes	Yes	No	Yes	Yes
Year $\times$ Province	No	No	Yes	No	No	Yes
Observations	10,951	10,872	10,872	10,993	10,905	10,905
Ineligible Group Mean	0.888	0.888	0.888	0.899	0.899	0.899
Adj R-squared	0.029	0.028	0.053	0.030	0.030	0.050

The dependent variable is the individual labor force participation. Variables of interest is the spouse's or own pension eligibility. Specifications (2)(3)(5)(6) include own and spouse's years of education. Standard errors in parentheses, clustered at the individual level. * $p < .1$, ** $p < .05$, *** $p < .01$.

We also conduct a placebo RDD at women's mandatory retirement age (50) in Table 13. We estimate a small decline in wives' participation and a small increase in husbands participation, neither statistically significant. The suggests an added-worker response on both sides, yet concentrated at husbands' eligibility, aligning with husbands' primary-earner role. It is also plausible that most rural women work in agriculture, less constrained by age-based rules (Giles et al., 2023), so mandatory retirement from formal jobs at 50 has limited bite for them.

Because the effects above arise primarily at the husbands' pension cutoff, we center the analysis on this treatment rather than on wives' eligibility. Accordingly, the subsequent results use a single-cutoff RDD at the husband's age-60 threshold. In Table 3, sectoral patterns are consistent with this focus: wives' participation rises by 5.1 p.p. ($p < 0.05$) in agriculture at husbands' eligibility, with no significant changes in other sectors, while husbands' non-agricultural employment falls by 5.2 p.p. This identifies non-agricultural jobs as the margin of loss at retirement and agriculture as the primary margin of adjustment for wives.

Table 3: Single regression discontinuity design on couples' labor force participation by sector

Outcome by Sector	Non Agriculture Employed	Home Business/Self Employed	Agriculture
Panel A: Women's Labor Force Participation			
Spouse's pension eligibility	-0.022 (0.03)	0.015 (0.02)	0.051** (0.02)
Controls	Yes	Yes	Yes
Year $\times$ Province	Yes	Yes	Yes
Observations	6,318	6,177	10,954
Ineligible Group Mean	0.253	0.170	0.674
Adj R-squared	0.034	0.025	0.043
Panel B: Men's Labor Force Participation			
Own pension eligibility	-0.052 (0.03)	-0.030 (0.03)	0.009 (0.02)
Controls	Yes	Yes	Yes
Year $\times$ Province	Yes	Yes	Yes
Observations	6,760	6,055	10,903
Ineligible Group Mean	0.561	0.265	0.696
Adj R-squared	0.072	0.028	0.040

The dependent variable is the individual labor force participation by sector. Variable of interest is the husband's pension eligibility. Standard errors are reported in parentheses and clustered at the individual level. * $p < .1$, ** $p < .05$, *** $p < .01$.

6.2 The Mechanism in Action: Sectoral Reallocation

Having identified the sectoral margin, we now ask how couples reallocate labor across subsample. If households insure against the income shock from formal job loss, we should see shifts out of formal work and into home production (agriculture and family business). Accordingly, we focus on the group where adjustments are most likely, defined by husbands who ever held a non-agricultural employed job before age 60,¹⁷ consistent with Propositions 1 and Corollary 1.

Table 4 supports this prediction, showing sectoral participation at the husbands age-60 cutoff. Non-agricultural participation falls by 7.1 p.p. ($p < 0.10$) while agricultural participation rises by 8.0 p.p. ($p < 0.05$), consistent with Corollary 1 and a shift from formal employment toward own production. At the same threshold, wives agricultural participation increases by 7.6 p.p. ($p < 0.10$), with no significant change in their non-agricultural or home-business participation. This coordinated reallocation within the household aligns with Proposition 1, with agriculture serving as the primary margin of adjustment.

¹⁷Our "ever non-agr job" split yields a much smaller sample because respondents reported non-agr employment far less often than agricultural work in the 2011, 2013, and 2015 surveys as you can see from the number of observations across sectors in Table 3.

Table 4: Single regression discontinuity design on couples’ labor force participation: husbands ever non-agricultural pre-60

Outcome by Sector	Non Agriculture Employed	Home Business/Self Employed	Agriculture
Panel A: Women’s Labor Force Participation			
Spouse’s pension eligibility	-0.011 (0.04)	0.016 (0.02)	0.076* (0.04)
Controls	Yes	Yes	Yes
Year $\times$ Province	Yes	Yes	Yes
Observations	2,253	2,109	3,645
Ineligible Group Mean	0.316	0.107	0.671
Adj R-squared	0.126	0.106	0.048
Panel B: Men’s Labor Force Participation			
Own pension eligibility	-0.071* (0.04)	0.021 (0.03)	0.080** (0.04)
Controls	Yes	Yes	Yes
Year $\times$ Province	Yes	Yes	Yes
Observations	2,962	2,170	3,727
Ineligible Group Mean	0.812	0.134	0.665
Adj R-squared	0.124	0.066	0.053

The dependent variable is the individual labor force participation. Variable of interest is husband’s pension eligibility. Standard errors are reported in parentheses and clustered at the individual level. Our “ever non-agr job” split yields a much smaller sample because respondents reported non-agr employment far less often than agricultural work in the 2011, 2013, and 2015 surveys. * $p < .1$, ** $p < .05$, *** $p < .01$.

By contrast, in the “never non-ag pre-60” subsample (Table 5, Panel A) we find no significant spousal response. The apparent 19.1 p.p. rise in husbands’ non-agricultural participation at their own eligibility is purely descriptive: the outcome is identically zero to the left of the age-60 cutoff, violating RDD design and mechanically producing a jump once a few men take non-agricultural jobs after 60. We therefore do not interpret this as a causal RDD effect, and it does not contradict the insurance mechanism. The same caveat applies to the change in non-agricultural employment income in Table 8, Panel B.

6.3 Time allocation and home production

Given the larger impacts among couples with husbands likely in formal employment before retirement, we focus next on this subgroup’s response to husbands’ pension eligibility. Table 6 reports sectoral time adjustments that mirror the participation results. At eligibility, husbands weekly non-agricultural hours decline by about 26.5 percent, while time on their own farm rises by roughly 33.8 percent ($p < 0.05$). Wives increase time on their own farm by about 27.4 percent ($p < 0.10$) with no detectable change in non-agricultural hours.¹⁸ This pattern where husbands exiting formal work and shifting into own production as wives expand farm time corroborates the sectoral reallocation in Table 4.

¹⁸The wife’s weekly hours spent on own farm before cutoff is 6.8 hrs/week, husband’s weekly hours before cutoff is 7 hrs/week.

Table 5: Single regression discontinuity design on couples' labor force participation: husbands never non-agricultural pre-60

Outcome by Sector	Non Agriculture Employed	Home Business/Self Employed	Agriculture
Panel A: Women's Labor Force Participation			
Spouse's pension eligibility	0.054 (0.05)	0.069 (0.05)	0.065 (0.06)
Controls	Yes	Yes	Yes
Year $\times$ Province	Yes	Yes	Yes
Observations	1,283	1,312	2,058
Ineligible Group Mean	0.150	0.258	0.657
Adj R-squared	0.100	0.149	0.067
Panel B: Men's Labor Force Participation			
Own pension eligibility	0.191*** (0.04)	0.058 (0.05)	0.021 (0.06)
Controls	Yes	Yes	Yes
Year $\times$ Province	Yes	Yes	Yes
Observations	1,354	1,477	2,092
Ineligible Group Mean	0.000	0.408	0.658
Adj R-squared	0.162	0.176	0.118

The dependent variable is the individual labor force participation. Variable of interest is husband's pension eligibility. Standard errors are reported in parentheses and clustered at the individual level. Our "ever non-agr job" split yields a much smaller sample because respondents reported non-agr employment far less often than agricultural work in the 2011, 2013, and 2015 surveys. * $p < .1$, ** $p < .05$, *** $p < .01$.

Table 7 documents shifts in family care at the intensive margin. Husbands substantially increase time caring for grandchildren (38.4 percent, $p < 0.05$) and for elderly parents (138.2 percent, $p < 0.01$). Wives reduce time caring for parents by 40.2 percent and increase time caring for grandchildren by 6.9 percent. Both are not statistically significant. This asymmetry runs counter to leisure-complementarity predictions that joint retirement equalizes household duties (Stancanelli and Soest, 2012). Instead, husbands added care reflects a reallocation toward home production following exit from formal employment, while wives maintain existing care responsibilities alongside increased agricultural work.

These time allocation patterns have two key implications. First, they rule out joint leisure as the main mechanism in this context. No effect on leisure activities is found for this group in Table 14 in appendix.¹⁹ Second, they reveal that retirement does not reduce wives' total work burden; rather, wives are deployed again to support home production, confirming the added-worker interpretation of our main results.

¹⁹Because the survey does not record exact leisure hours, we proxy leisure using items from the CHARLS Lifestyle and Health Behaviors module (noting that in 2011 this module lacks questions on the purpose of physical activity). We found significant increase in leisure activities for the couple whose husbands never work in non-agricultural sector from Table 15.

Table 6: Single regression discontinuity design on weekly work time for couples with husbands ever non-agricultural pre-60

Outcome by Sector	Non Agriculture	Agriculture
Panel A: Women's Time Spent on Work		
Spouse's pension eligibility	-0.003 (0.20)	0.274* (0.16)
Controls	Yes	Yes
Year $\times$ Province	Yes	Yes
Observations	2,114	3,597
Ineligible Group Mean	1.478	2.616
Adj R-squared	0.131	0.080
Panel B: Men's Time Spent on Work		
Own pension eligibility	-0.265 (0.19)	0.338** (0.16)
Controls	Yes	Yes
Year $\times$ Province	Yes	Yes
Observations	2,729	3,687
Ineligible Group Mean	3.881	2.644
Adj R-squared	0.131	0.075

The dependent variable is the inverse hyperbolic sine of individual time spent per week on work by sector (non agricultural employed work and own farm agricultural work). Variable of interest is husband's pension eligibility. Standard errors are reported in parentheses and clustered at the individual level. * $p < .1$, ** $p < .05$, *** $p < .01$.

6.4 Income effects and consumption responses

The labor reallocation documented above should translate into observable changes in household income. Table 8 directly test whether the within-household reallocation offsets the income shock from husbands' job loss. The top panel restricted to husbands more likely to retire from formal employment, confirms the mechanism: monthly non-agricultural employment earnings fall sharply at own eligibility by 80.6 percent ($p < 0.05$). Because coefficients under the inverse hyperbolic sine transformation can possibly exceed one, we also report an interpretable average treatment effect for non-agricultural employment income and home business income following previous literature (Chen and Roth, 2023): $\theta_{ATE\%} = \frac{E[Y^{\text{post}} - Y^{\text{pre}}]}{E[Y^{\text{pre}}]}$. By this measure, reaching eligibility reduces husbands' non-agricultural employment earnings by about 29.5 percent relative to the control mean.

Table 7: Single regression discontinuity design on home care for couples with husbands ever non-agricultural pre-60

Outcome by Care Recipient	Grandchildren	Parents
Panel A: Women's Time Spent on Home Care		
Spouse's pension eligibility	0.069 (0.23)	-0.402 (0.29)
Controls	Yes	Yes
Year $\times$ Province	Yes	Yes
Observations	2,248	383
Ineligible Group Mean	1.664	0.992
Adj R-squared	0.079	0.551
Panel B: Men's Time Spent on Home Care		
Own pension eligibility	0.384** (0.19)	1.382*** (0.42)
Controls	Yes	Yes
Year $\times$ Province	Yes	Yes
Observations	2,187	239
Ineligible Group Mean	0.971	0.963
Adj R-squared	0.096	0.454

The dependent variable is the inverse hyperbolic sine of individual time spent on home care. Variable of interest is husband's pension eligibility. Standard errors are reported in parentheses and clustered at the individual level. * $p < .1$, ** $p < .05$, *** $p < .01$.

On the offsetting side, household agricultural income and home business income both rise at husbands' eligibility by roughly 12.3 percent and 24.1 percent, consistent with couples reallocating toward home production documented in Tables 4-7,²⁰ though both of them are not statistically significant.

By contrast, the bottom panel shows that, among husbands less likely to retire from formal jobs, eligibility significantly reduces household agricultural income by 58.7 percent ($p < 0.05$). This pattern suggests Type L households rely more heavily on own-farm production at baseline, and pension eligibility allows them to reduce this labor-intensive activity, consistent with a pure positive income effect from the transfer. Although their main job income appears to rise at eligibility, this is driven by a near-zero baseline and is not at odds with our model's predictions. Taken together with the sectoral and time use results, the evidence points to a classic added-worker response on the wife's side and a formal to informal shift on the husband's side.

²⁰Because the questionnaire revised the questions on agricultural participation in the agricultural income section in 2013 and 2015, we focus on the intensive margin for agricultural production revenue. Therefore we directly report the coefficients for this outcome in Table 3.

Table 8: Single regression discontinuity design on husbands' monthly income

Monthly Income	Non-Agr Employment	Agr Production	Home Business
Panel A: Ever Non-agr			
Spouse's pension eligibility	-0.806** (0.35)	0.123 (0.20)	0.241 (0.16)
Controls	Yes	Yes	Yes
Year $\times$ Province	Yes	Yes	Yes
Observations	3,410	1,495	3,682
Ineligible Group Mean	5.409	6.629	0.407
Adj R-squared	0.051	0.076	0.010
Panel B: Never Non-agr			
Own pension eligibility	1.134*** (0.31)	-0.587** (0.30)	0.224 (0.30)
Controls	Yes	Yes	Yes
Year $\times$ Province	Yes	Yes	Yes
Observations	1,592	915	2,047
Ineligible Group Mean	0.311	7.173	1.381
Adj R-squared	0.075	0.058	0.037

The dependent variable is the inverse hyperbolic sine of monthly income from employment, household agricultural production, and home business. Variable of interest is husband's pension eligibility. Standard errors are reported in parentheses and clustered at the individual level. * $p < .1$, ** $p < .05$, *** $p < .01$.

From what we have done above, we uncover a clear spousal-spillover mechanism. At the husbands' cutoff, wives' labor force participation rises by 4 to 5 p.p. on average and the increase is concentrated in agriculture. Husbands exit non-agricultural work and reallocate toward agriculture, home business/self-employment, and caregiving, with own-production revenues rising precisely where the income loss from formal-job retirement is larger. In households whose husbands ever work in non-agricultural sector, the 7.6 p.p. increase in wives' agricultural participation offsets husbands' decline in non-agricultural participation at the extensive margin, with hours moving in the same direction. These magnitudes align with the model under substitutable non-market time: retirement creates an insurance problem that households address through intra-household reallocation.

Table 9: Double regression discontinuity design on women’s household weekly consumption

Outcome by Consumption Type	Food consumption Per Capita	Alcohol/Cigarette consumption
Spouse’s pension eligibility	-0.035 (0.05)	-0.101 (0.13)
Own pension eligibility	0.085* (0.05)	0.274** (0.13)
Controls	Yes	Yes
Year $\times$ Province	Yes	Yes
Observations	10,162	10,452
Ineligible Group Mean	4.972	2.701
Adj R-squared	0.096	0.034

The dependent variables are the inverse hyperbolic sine of household food consumption per capita and household alcohol/cigarette consumption. Variables of interest are the spouse’s and own pension eligibility. Standard errors are reported in parentheses and clustered at the individual level. * $p < .1$, ** $p < .05$, *** $p < .01$.

Consistent with Proposition 2 serving as a test of the pure transfer effect, wives’ own eligibility operates mainly as a transfer rather than a change in labor market access. Most rural women have already exited formal employment even earlier than age 50, so reaching age 60 leaves their labor supply essentially unchanged (Tables 2, 11) but raises household consumption: food spending per capita increases by 8.5 percent ($p < 0.10$) and alcohol/cigarette spending by 27.4 percent ($p < 0.05$) in Table 9. In comparison, the husband’s eligibility produces large labor responses—combining a negative wage shock from formal-job loss with a modest transfer while generating no comparable consumption bump for the couple. The asymmetry strengthens our interpretation that the driving mechanism is income loss at the husband’s cutoff, met by added work from the wife, whereas the wife’s cutoff delivers a pure transfer that increases consumption without shifting labor.

7 Further Discussions

This section delivers two indicators and interpret them connecting to the results reported in the previous sections:

- (a) a *household level welfare-change lower bound* based on the income change ΔY ;
- (b) a *income shock offsetting decomposition* that quantifies how the husbands earnings loss is covered by the pension and by each spouses reallocation into household production.

7.1 Welfare Implications of Access Loss

Let $Y(s) = w(s) + r + B(s)$ be full income and let $\Pi(w(s), r)$ be the CES dual. With indirect utility in (3), the compensating variation (CV) that keeps pre-state utility at post-state prices/income solves

$$(\alpha + \phi) \log(Y_{\text{post}} + \text{CV}) - \phi \log \Pi_{\text{post}} = (\alpha + \phi) \log Y_{\text{pre}} - \phi \log \Pi_{\text{pre}},$$

hence

$$CV = Y_{\text{pre}} \left(\frac{\Pi_{\text{post}}}{\Pi_{\text{pre}}} \right)^{\frac{\phi}{\alpha+\phi}} - Y_{\text{post}}, \quad \Pi_{\text{pre}} = \Pi(\bar{w}, r), \quad \Pi_{\text{post}} = \Pi(r, r). \quad (15)$$

The equivalent variation (EV) is

$$EV = Y_{\text{pre}} - Y_{\text{post}} \left(\frac{\Pi_{\text{pre}}}{\Pi_{\text{post}}} \right)^{\frac{\phi}{\alpha+\phi}}. \quad (16)$$

When the husband loses access to formal jobs, non-market time becomes relatively cheaper (lower shadow price), so the household substitutes toward G , partially offsetting the utility impact of lower full income.

Under access loss, $\Pi_{\text{post}} \leq \Pi_{\text{pre}}$ so (15) implies

$$CV \leq Y_{\text{pre}} - Y_{\text{post}}. \quad (17)$$

An identical bound holds for EV. Thus the *income loss is an upper bound on the money-metric welfare loss*. Using the RD estimates, define the reduced-form jump in full income at the cutoff

$$\widehat{\Delta Y} \equiv \widehat{\Delta y}_{\text{nonag wage}} + \widehat{\Delta y}_{\text{ag prod}} + \widehat{\Delta y}_{\text{home biz}} + \widehat{\Delta B}.$$

With $\Delta Y = Y_{\text{post}} - Y_{\text{pre}}$, the bound becomes

$$\widehat{CV} \leq (-\widehat{\Delta Y})^+, \quad (x)^+ \equiv \max\{x, 0\}. \quad (18)$$

Calibrated CV interval. If (σ, γ) (or their bounds) are known, let $t(\sigma, \gamma) \equiv (\Pi_{\text{post}}/\Pi_{\text{pre}})^{\phi/(\alpha+\phi)} \in (0, 1]$. Then $CV = (Y_{\text{pre}} - Y_{\text{post}}) + Y_{\text{pre}}(t - 1)$ and for $t \in [t_L, t_U]$:

$$CV \in \left[(Y_{\text{pre}} - Y_{\text{post}}) + Y_{\text{pre}}(t_U - 1), (Y_{\text{pre}} - Y_{\text{post}}) + Y_{\text{pre}}(t_L - 1) \right].$$

These income changes provide conservative (worst-case) bounds on welfare effects. The actual welfare losses are smaller because households successfully reallocate labor, as documented in Tables 4-7.

7.2 Shock-response Decompositions

The husband's loss of formal employment at age 60 creates a substantial income shock. How do households insure against this loss? While our unitary model cannot decompose welfare effects across spouses, we can analyze how the earnings shortfall is offset through observable income channels. This decomposition reveals the quantitative importance of wives' labor supply response documented in our main results.

Framework. At the husband’s eligibility threshold, his formal (main-job) income falls by $|\Delta y_h^F| > 0$, where Δ denotes the RD jump at age 60. This loss can be offset through three potential channels: (i) the pension benefit B , (ii) increased household agricultural income Δy^A , and (iii) increased home business income Δy^{HB} . We define the coverage ratio as:

$$\kappa = \frac{B + \Delta y^A + \Delta y^{HB}}{S}$$

representing the fraction of formal income loss offset through these channels, with residual shortfall $1 - \kappa$.

Allocating agricultural income across spouses. The household agricultural income increase reflects both spouses’ added hours. To attribute this income across spouses, we assume that within the household farm, spouses face a common marginal return to labor (efficient allocation under pooled resources). Under this assumption, each spouse’s contribution to agricultural income is proportional to their added hours. Using estimates from Table 6, wives contribute 44.1% of added agricultural hours (1.8632 vs 2.366 hours/week), implying they generate the same fraction of added agricultural income.²¹

Welfare Bound, Decomposition and Interpretation. Using RD estimates for households where the husband ever worked in non-agriculture, we decompose how they absorb the income shock at the age-60 cutoff. Applying the inverse-asinh transformation as in Chen and Roth (2023) to the pre-cutoff monthly baselines (1,912.29 yuan for non-agricultural employment, 378.50 yuan for agriculture income, and 122.11 yuan for home business earnings), we obtain the following changes: non-agricultural income falls by 29.48% (-563.74 yuan/month) compared to pre-60 employed income, while agricultural income rises by 12.30% (+46.56 yuan) and home business income rises by 50.64% (+61.84 yuan). Including the pension transfer (+120.60 yuan), the total offset is +229.00 yuan/month. The residual income gap of 334.75 yuan/month provides a credible upper bound on the household’s money-metric welfare loss by equation (17).

These income changes provide conservative (worst-case) bounds on welfare effects. The money-metric welfare loss is bounded above by the net fall in full income (equation (18)); the actual welfare loss is smaller because households substitute toward home production when formal access is lost (Tables 4-7).

Table 10 shows that the offset coming from reallocated labor is almost the same amount as pension benefit. For agricultural income, we allocate shares across spouses based on their time contributions; for home business, while we lack gender-specific time data, the income increase reflects joint household reallocation by both spouses into this sector. The pension provides 21.4% of the offset. Overall, household labor reallocation (agriculture and home business combined) covers

²¹This follows directly from linear technology: with common marginal return r^A , we have $\Delta y^A = r^A(\Delta I_w + \Delta I_h)$, so the wife’s contribution is $r^A \Delta I_w = \lambda_w \cdot \Delta y^A$ where $\lambda_w = \frac{\Delta I_w}{\Delta I_w + \Delta I_h}$.

19.2% of the loss, rising to 40.6% with the pension.²²

Table 10: Decomposition of the Income Shock Offset (Ever non-agr Group)

Channel	Yuan/month	% of non-agr loss
Agricultural income (total)	+46.56	8.3%
<i>Wife's share</i> ($\lambda_w \approx 44.1\%$)	+20.53	3.64%
<i>Husband's share</i> ($1 - \lambda_w \approx 55.9\%$)	+26.03	4.66%
Home business income (household)	+61.84	11.0%
Pension benefit	+120.60	21.4%
Total offset	+229.00	40.6%
Residual income gap	334.75	59.4%

Notes: Percentages in the third column are relative to the nonagricultural income loss.

While households mitigate the shock through substitution into home production, this reallocation itself carries hidden costs. The wife's increased labor contribution preserves household consumption by converting non-market time into productive work, a welfare tradeoff cannot be revealed on an aggregate level.

8 Conclusion

This paper documents a reversal of the joint-retirement pattern commonly observed in richer economies. In a setting with low replacement rates of pension on pre-retirement income and rigid age-based hiring rules, a husband's pension eligibility *increases* his wife's labor supply. A distinctive feature of our setting is the staggered exit-and-eligibility ages (male 60; female 50/60). This institutional timing furnishes a rare design leverage to *separate channels*: at the husband's cutoff, we observe an access-loss plus small-transfer shock, whereas at the wife's cutoff we observe a *transfer-only* shock. This separation allows us to meaningfully discuss spousal reallocation to the price component of retirement.

Using a regression discontinuity design on rural pension enrollees in China, we find that when a husband reaches the age-60 threshold, his wife's labor force participation rises by 5.3-5.4 percentage points. Inversely, husband's participation does not respond at wife's eligibility. This asymmetry underscores the role of institutions in shaping couples' behavior and aligns with evidence of stronger husband-to-wife spillovers at pension eligibility.

Our analysis indicates that the result is driven by an income-insurance mechanism within couples. At the husband's eligibility age, access to formal non-agricultural work effectively ends, creating a negative earnings shock that the pension does not fully replace. In a low-income, low-benefit setting, couples respond by raising labor supply to smooth consumption rather than relying

²²This decomposition analyzes income flows, not welfare distribution. In our unitary model; The income shares presented focus on the monetary scope for offsetting income loss.

solely on pension income. We document three facts that support this mechanism. First, the spousal response is concentrated among households whose husbands ever work in non-agricultural sector, where income loss is larger; in this group, wives' participation rises by 7.6 percentage points. Second, the adjustment operates through sectoral reallocation: husbands move out of formal work and into agriculture and home production, while wives increase agricultural labor. Third, the effect is asymmetric: a wife's own eligibility does not elicit a comparable labor response from her husband and instead functions as a small transfer that modestly raises household consumption. Together, these patterns point to the loss of the husband's formal job opportunities as the primary driver of couples' labor reallocation, not age or leisure complementarities.

These findings have important implications for pension reform in China and other low and middle-income countries. Behavioral responses to retirement income incentives often occur at the household level. Ignoring this interdependence, i.e., treating individuals as isolated decision-makers, can lead to misguided predictions and suboptimal policy. There are discussions in recent years in China about raising statutory retirement ages to 63 for men and 55/58 for women (Fang and Feng, 2018; Deng et al., 2023). Our findings suggest that this could directly improve wives' welfare by delaying the employment barrier that compels spousal labor increases. However, our results emphasize that protecting household welfare requires more than extended working lives. The same concerns apply more broadly across the developing world, where over two billion people face old age with pensions below subsistence (Barrientos et al., 2003; Cai et al., 2012; Giles et al., 2011). When pension systems fail to account for low replacement rates and rigid labor markets, they inadvertently shift old-age support burdens onto spouses, creating gendered consequences and turning retirement into a household financial crisis. Effective reform, therefore, requires a comprehensive approach: higher replacement rates, flexible employment for older workers, and coordination with social assistance to genuinely protect vulnerable families.

Our analysis points out several paths for future work. A collective household model could unpack how retirement shocks shift bargaining power and resource allocation, enabling richer welfare analysis beyond the unitary framework. More broadly, we bridge evidence from developing and high-income settings by showing that spousal labor as insurance versus joint leisure is not a fixed cultural trait but a rational response to incentives. This points to a tipping point in pension generosity: below it, benefits may intensify financial stress and trigger spousal substitution; above it, they can support coordinated, dignified retirement. A key design question follows: at what replacement rate do benefits begin to encourage joint retirement rather than the substitution we document? Identifying that threshold will require leveraging variation in pension generosity across places and over time.

A Model details and proofs

A.1 Glossary of Notations

Glossary of Notations	
Symbol	Meaning
$i \in \{h, w\}$	husband / wife
$s \in \{\text{pre}, \text{post}\}$	before vs. after eligibility
F_i, I_i, S_i	formal, informal/agricultural, non-market time
$L_i = 1 - S_i$	market time
$r > 0$	informal/agricultural return (both spouses)
$\bar{w} > r$	husband's formal return when accessible
$w(s) \in \{\bar{w}, r\}$	husband's accessible return in state s
$B(s) = B \cdot \mathbf{1}\{s = \text{post}\}$	pension transfer post eligibility
$U = \alpha \log C + \phi \log G(S_h, S_w)$	preferences
$G = [\gamma S_h^\rho + (1 - \gamma) S_w^\rho]^{1/\rho}$	CES non-market aggregator
$\sigma = \frac{1}{1-\rho} > 0$	elasticity of substitution in G
$\Pi(w(s), r)$	CES unit cost of one unit of G (derived below)
$Y(s) = w(s) + r + B(s)$	full income

A.2 Deriving the CES unit cost and Hicksian time demands

We solve the unit-expenditure problem

$$\min_{S_h, S_w} \{ w S_h + r S_w : [\gamma S_h^\rho + (1 - \gamma) S_w^\rho]^{1/\rho} \geq 1 \}.$$

Let λ be the multiplier. First-order conditions imply

$$w = \lambda \gamma S_h^{\rho-1} (\gamma S_h^\rho + (1 - \gamma) S_w^\rho)^{\frac{1}{\rho}-1}, \quad r = \lambda (1 - \gamma) S_w^{\rho-1} (\gamma S_h^\rho + (1 - \gamma) S_w^\rho)^{\frac{1}{\rho}-1}.$$

Taking ratios,

$$\frac{S_h}{S_w} = \left(\frac{\gamma}{1 - \gamma} \right)^\sigma \left(\frac{w}{r} \right)^{-\sigma}.$$

Plugging into the aggregator at the optimum and using homogeneity yields the unit cost

$$\Pi(w(s), r) = \left[\gamma^\sigma w^{1-\sigma} + (1 - \gamma)^\sigma r^{1-\sigma} \right]^{\frac{1}{1-\sigma}}.$$

Hicksian time demands per unit of G are

$$S_h^H(1) = \gamma^\sigma \Pi(w(s), r)^\sigma w^{-\sigma}, \quad S_w^H(1) = (1 - \gamma)^\sigma \Pi(w(s), r)^\sigma r^{-\sigma}.$$

Combining with $G^*(s) = \frac{\phi}{\alpha + \phi} \frac{Y(s)}{\Pi(w(s), r)}$ (Cobb-Douglas in the top tier) yields the Marshallian $S_i^*(s)$ in (2).

A.3 Closed-form allocations and indirect utility

Equation (1) and the Hicksian shares deliver

$$C^*(s) = \frac{\alpha}{\alpha + \phi} Y(s), \quad S_h^*(s) = \frac{\phi}{\alpha + \phi} Y(s) \frac{\gamma^\sigma}{\Pi(w(s), r)^{1-\sigma} w(s)^\sigma}, \quad S_w^*(s) = \frac{\phi}{\alpha + \phi} Y(s) \frac{(1 - \gamma)^\sigma}{\Pi(w(s), r)^{1-\sigma} r^\sigma},$$

and $L_i^*(s) = 1 - S_i^*(s)$. Indirect utility is (3).

A.4 Proof of Proposition 1 (threshold form)

In Type H, $w(\text{pre}) = \bar{w}$, $w(\text{post}) = r$. Using (2), the wives non-market time satisfies

$$\frac{S_w^{\text{post}}}{S_w^{\text{pre}}} = \frac{Y_{\text{post}}}{Y_{\text{pre}}} \left(\frac{\Pi(\bar{w}, r)}{\Pi(r, r)} \right)^{1-\sigma} = \frac{Y_{\text{post}}}{Y_{\text{pre}}} \left(\frac{\Pi(r, r)}{\Pi(\bar{w}, r)} \right)^{\sigma-1}.$$

Because $L_w = 1 - S_w$, we have $\Delta L_w > 0 \iff S_w^{\text{post}} < S_w^{\text{pre}}$:

$$\frac{Y_{\text{post}}}{Y_{\text{pre}}} \left(\frac{\Pi(r, r)}{\Pi(\bar{w}, r)} \right)^{\sigma-1} < 1.$$

Normalize by r : let $g = \bar{w}/r > 1$, $R = B/\bar{w} \in [0, 1]$. Then $Y_{\text{pre}}/r = g + 1$, $Y_{\text{post}}/r = 2 + gR$, and

$$\Pi(r, r)^{1-\sigma} = \gamma^\sigma + (1 - \gamma)^\sigma =: A, \quad \Pi(w(s), r)^{1-\sigma} = \gamma^\sigma g^{1-\sigma} + (1 - \gamma)^\sigma =: D.$$

The inequality becomes

$$\frac{2 + gR}{g + 1} < \frac{A}{D} \iff R < \underbrace{\frac{g + 1}{g} \cdot \frac{A}{D} - \frac{2}{g}}_{=: R^*(\sigma; g, \gamma)}.$$

This yields the if-and-only-if statement with explicit threshold $R^*(\sigma; g, \gamma)$.

A.5 Monotonicities of R^*

Increasing in g . Write $R^* = \frac{g+1}{g} \frac{A}{D} - \frac{2}{g}$ with $A = \gamma^\sigma + (1 - \gamma)^\sigma$ and $D = \gamma^\sigma g^{1-\sigma} + (1 - \gamma)^\sigma$. Differentiation gives

$$\frac{\partial R^*}{\partial g} = \frac{2 - \frac{A}{D}}{g^2} + \left(1 + \frac{1}{g} \right) \frac{A \gamma^\sigma (\sigma - 1) g^{-\sigma}}{D^2}.$$

For all $g > 1$, $\sigma \geq 1$, and $\gamma \in (0, 1)$ the right-hand side is strictly positive (details as in the working notes), so R^* is increasing in g .

Increasing in σ under balanced weights. For $\gamma = \frac{1}{2}$, one obtains $\frac{A}{D} = \frac{2}{1+g^{1-\sigma}}$ and

$$R^*(\sigma; g, \tfrac{1}{2}) = \frac{2}{g} \cdot \frac{g - g^{1-\sigma}}{1 + g^{1-\sigma}}.$$

Let $t = g^{1-\sigma}$ (so $dt/d\sigma = -\ln g \cdot t < 0$). Then

$$\frac{\partial R^*}{\partial \sigma} = \frac{2}{g} \cdot \frac{g+1}{(1+t)^2} \cdot (\ln g) t > 0.$$

For generic $\gamma \in (0, 1)$, $\partial R^*/\partial \sigma$ is not sign-definite globally; the balanced-weight case (and a neighborhood thereof) delivers the clean monotonicity used in the text.

A.6 Proof of Corollary 1

Linear returns with access constraints imply specialization: once the higher-return formal opportunity is infeasible, the husbands formal hours are zero; any market time flows to informal/agriculture. The wives additional market time (when it occurs) must also be in informal/agriculture where access does not bind.

A.7 Proof of Proposition 2

If $w(\text{post}) = w(\text{pre})$ and the wives accessible return is r in both states, then $\Pi_{\text{post}} = \Pi_{\text{pre}}$. From (1), C^* rises by $\frac{\alpha}{\alpha+\phi}B$; from (2), S_i^* rises proportionally with Y , so L_i^* falls for $i \in \{h, w\}$. Because the price ratio w/r is unchanged, the intra- G split is unchanged.

A.8 Comparative statics in ϕ

From (2), $S_w^*(s) = \left(\frac{\phi}{\alpha+\phi}\right) \cdot Y(s) \cdot \kappa(s)$ with $\kappa(s)$ independent of ϕ . Therefore

$$\frac{S_w^{\text{post}}}{S_w^{\text{pre}}} = \frac{Y_{\text{post}}}{Y_{\text{pre}}} \left(\frac{\Pi(r, r)}{\Pi(\bar{w}, r)} \right)^{1-\sigma},$$

and the threshold condition in Proposition 1 is *independent of ϕ* . However, response magnitudes scale with ϕ :

$$\frac{\partial S_w^*(s)}{\partial Y} = \frac{\phi}{\alpha + \phi} \cdot \kappa(s), \quad \frac{\partial S_w^*(s)}{\partial \log \Pi} = -(1 - \sigma) S_w^*(s).$$

Thus a larger ϕ makes the household more sensitive to any given change in Y or Π (amplification). By CobbDouglas at the top tier, the marginal propensity to consume out of a transfer equals $\text{MPC}_C = \frac{\alpha}{\alpha+\phi}$, which *decreases* in ϕ .

A.9 Money-metric welfare: CV/EV and the upper bound

Indirect utility (3) implies the CV/EV formulas in (15). If access loss reduces the composite price ($\Pi_{\text{post}} \leq \Pi_{\text{pre}}$), then

$$\text{CV} - (Y_{\text{pre}} - Y_{\text{post}}) = Y_{\text{pre}} \left[\left(\frac{\Pi_{\text{post}}}{\Pi_{\text{pre}}} \right)^{\frac{\phi}{\alpha+\phi}} - 1 \right] \leq 0,$$

hence $0 \leq \text{CV} \leq Y_{\text{pre}} - Y_{\text{post}}$.

A.10 From inverse-hyperbolic-sine coefficients to level jumps

Let $y \geq 0$ be an income component and suppose the RDD is estimated in $\text{asinh}(y)$:

$$\text{asinh}(y) = \log(y + \sqrt{y^2 + 1}).$$

Let $\bar{y}_0$ be the local mean just below the cutoff and $\hat{\beta}$ the estimated jump in $\text{asinh}(y)$. A consistent level-scale jump is

$$\widehat{\Delta y} = \sinh(\text{asinh}(\bar{y}_0) + \hat{\beta}) - \bar{y}_0.$$

Aggregating across components $k \in \{\text{non-ag wage, ag production, home business, pension}\}$ yields $\widehat{\Delta Y} = \sum_k \widehat{\Delta y}_k$ for use in the conservative welfare bound in Section 7.1.

B Two-Period Life-Cycle Extension

We embed the static pre/post framework in a parsimonious two-period model to clarify how intertemporal smoothing and liquidity shape the spousal response at the husbands eligibility cutoff. Periods are $t \in \{0, 1\}$ with eligibility at $t = 1$.

Preferences and aggregator. Preferences retain the Cobb-Douglas in (C_t, G_t) and the intra- G CES across spouses:

$$\max_{\{C_t, G_t, S_{i,t}, a_{t+1}\}} \sum_{t=0}^1 \beta^t [\alpha \log C_t + \phi \log G_t], \quad G_t = [\gamma S_{h,t}^\rho + (1 - \gamma) S_{w,t}^\rho]^{1/\rho}, \quad \sigma = \frac{1}{1-\rho} > 0.$$

Time: $F_{i,t} + I_{i,t} + S_{i,t} = 1$ for $i \in \{h, w\}$.

Wages, access, and unit cost. The husband loses formal access at eligibility: $w_{h,0} = \bar{w} > r$, $w_{h,1} = r$; the wifes accessible return is r in both periods. A pension $B \geq 0$ arrives at $t = 1$. The CES unit cost of G_t is

$$\Pi_t \equiv \Pi(w_{h,t}, r) = \left[\gamma^\sigma w_{h,t}^{1-\sigma} + (1 - \gamma)^\sigma r^{1-\sigma} \right]^{\frac{1}{1-\sigma}}, \quad \Pi_1 = \Pi(r, r) \leq \Pi_0 = \Pi(\bar{w}, r).$$

The key insight is that inter-temporal smoothing capacity shapes the spousal response: when households can freely borrow or save, they partially absorb the husband's income shock through the financial system, dampening the wife's labor adjustment. By contrast, when liquidity constraints bind, preventing borrowing against future pension income would amplify the added-worker effect. This framework thus delivers a condition that nests both the static threshold and predicts stronger spousal responses precisely when credit markets fail to insure retirement shocks.

Budgets and assets. Let a_t be assets at the start of t and $1 + r_f$ the gross interest factor. Using the full-income convention of the main text, period resources are

$$Y_t := w_{h,t} + r + B \cdot \mathbf{1}\{t = 1\}.$$

The budget with a borrowing limit $a_{t+1} \geq -\bar{b}$ and $a_2 \geq 0$ is

$$C_t + \Pi_t G_t + a_{t+1} = Y_t + (1 + r_f)a_t.$$

Let $E_t := C_t + \Pi_t G_t$ denote expenditure on (C_t, G_t) . Cobb-Douglas implies

$$C_t = \frac{\alpha}{\alpha + \phi} E_t, \quad G_t = \frac{\phi}{\alpha + \phi} \frac{E_t}{\Pi_t}, \quad E_t = \frac{\alpha + \phi}{\alpha} C_t.$$

Intertemporal margin. If the borrowing constraint is slack, the Euler condition yields

$$\frac{C_1}{C_0} = \beta(1 + r_f) \quad \implies \quad \frac{E_1}{E_0} = \beta(1 + r_f).$$

If the constraint binds at $t = 0$ (e.g., $a_1 = -\bar{b}$ with $\bar{b} = 0$), then $E_0 = Y_0$ and $E_1 = Y_1$, so $\frac{E_1}{E_0} = Y_1/Y_0$.

Proposition 3 (Dynamic added-worker condition). *The wife increases market time at eligibility ($\Delta L_w > 0$) iff*

$$\boxed{\frac{S_{w,1}}{S_{w,0}} = \underbrace{\frac{E_1}{E_0}}_{\kappa} \left(\frac{\Pi_1}{\Pi_0} \right)^{\sigma-1} < 1}.$$

Hence $\kappa = \beta(1 + r_f)$ when smoothing is unconstrained and $\kappa = Y_1/Y_0$ under a binding liquidity constraint (with no borrowing). Larger σ (greater substitutability across spouses nonmarket time) and larger formal premium $g = \bar{w}/r$ (which lowers Π_1/Π_0) both increase the likelihood of $\Delta L_w > 0$.

Proof. Hicksian demands for the CES lower tier imply $S_{w,t} = \frac{\phi}{\alpha + \phi} E_t \cdot \frac{(1-\gamma)^\sigma}{r^\sigma} \Pi_t^{\sigma-1}$, hence $S_{w,1}/S_{w,0} = (E_1/E_0)(\Pi_1/\Pi_0)^{\sigma-1}$. Since $L_{w,t} = 1 - S_{w,t}$, $\Delta L_w > 0$ iff $S_{w,1}/S_{w,0} < 1$. The expressions for E_1/E_0 follow from the Euler equation and the binding-constraint case. $\square$

Static model as a special case. If assets are fixed (or smoothing is infeasible), identify E_t with the static full income $Y(s)$ to recover the threshold as in the main text. Under zero borrowing

capacity and no assets, $E_t = Y(s_t)$ and $\kappa = Y(\text{post})/Y(\text{pre})$.

In particular, with $g \equiv \bar{w}/r$ and $R \equiv B/\bar{w}$,

$$\frac{E_1}{E_0} = \frac{Y(\text{post})}{Y(\text{pre})} = \frac{2 + gR}{g + 1}, \quad \Pi_0 = \Pi(\bar{w}, r), \quad \Pi_1 = \Pi(r, r),$$

so the condition $(S_{w,1}/S_{w,0} < 1)$ collapses exactly to $R < R^*(\sigma; g, \gamma)$ in Proposition 1.

C Additional Results & Robustness Check

Table 11: Double regression discontinuity design on couples' labor force participation

	(1)	(2)	(3)
Panel A: Women's Labor Force Participation			
Spouse's pension eligibility	0.045*	0.046*	0.044*
	(0.02)	(0.02)	(0.02)
Own pension eligibility	0.025	0.023	0.017
	(0.02)	(0.02)	(0.02)
Controls	No	Yes	Yes
Year $\times$ Province	No	No	Yes
Observations	10,820	10,741	10,739
Ineligible Group Mean	0.791	0.791	0.791
Adj R-squared	0.010	0.010	0.044
Panel B: Men's Labor Force Participation			
Spouse's pension eligibility	-0.021	-0.021	-0.021
	(0.02)	(0.02)	(0.02)
Own pension eligibility	-0.011	-0.009	-0.008
	(0.02)	(0.02)	(0.02)
Controls	No	Yes	Yes
Year $\times$ Province	No	No	Yes
Observations	10,765	10,692	10,692
Ineligible Group Mean	0.902	0.902	0.902
Adj R-squared	0.021	0.022	0.050

The dependent variable is the individual labor force participation. Variables of interest are the spouse's and own pension eligibility. Specifications (2) and (3) include own and spouse's years of education. Standard errors are reported in parentheses and clustered at the individual level. * $p < .1$, ** $p < .05$, *** $p < .01$.

Table 12: Double regression discontinuity design on couples' labor force participation

	(1)	(2)	(3)
Panel A: Women's Labor Force Participation			
Spouse's pension eligibility	0.041 (0.03)	0.042* (0.03)	0.044* (0.03)
Own pension eligibility	0.022 (0.02)	0.020 (0.02)	0.016 (0.02)
Controls	No	Yes	Yes
Year $\times$ Province	No	No	Yes
Observations	9,191	9,125	9,123
Ineligible Group Mean	0.788	0.788	0.788
Adj R-squared	0.010	0.010	0.042
Panel B: Men's Labor Force Participation			
Spouse's pension eligibility	-0.026 (0.02)	-0.026 (0.02)	-0.030 (0.02)
Own pension eligibility	-0.019 (0.02)	-0.017 (0.02)	-0.018 (0.02)
Controls	No	Yes	Yes
Year $\times$ Province	No	No	Yes
Observations	9,158	9,093	9,093
Ineligible Group Mean	0.901	0.901	0.901
Adj R-squared	0.023	0.023	0.050

The dependent variable is the individual labor force participation. Variables of interest are the spouse's and own pension eligibility. Specifications (2) and (3) include own and spouse's years of education. Same age couples are dropped in the regression for robustness check. Standard errors are reported in parentheses and clustered at the individual level. * $p < .1$, ** $p < .05$, *** $p < .01$.

Table 13: Single regression discontinuity design on couples' labor force participation at 50

	(1)	(2)	(3)
Panel A: Women's Labor Force Participation			
Own age 50	-0.039 (0.05)	-0.029 (0.05)	-0.036 (0.05)
Controls	No	Yes	Yes
Year $\times$ Province	No	No	Yes
Observations	3,454	3,398	3,392
Ineligible Group Mean	0.839	0.839	0.839
Adj R-squared	0.003	0.002	0.029
Panel B: Men's Labor Force Participation			
Spouse's age 50	0.030 (0.04)	0.028 (0.04)	0.016 (0.04)
Controls	No	Yes	Yes
Year $\times$ Province	No	No	Yes
Observations	3,416	3,373	3,368
Ineligible Group Mean	0.926	0.926	0.926
Adj R-squared	0.002	0.002	0.014

The dependent variable is the individual labor force participation. Variables of interest are the spouse's or own pension eligibility. Own age bracket and spouse's age bracket in this table are from 45 to 55. Specifications (2) and (3) include own and spouse's years of education. Standard errors are reported in parentheses and clustered at the individual level. * $p < .1$, ** $p < .05$, *** $p < .01$.

Table 14: Single regression discontinuity design on couples' leisure activities: husbands ever non-agricultural pre-60

	Have Leisure Activities	Day Spent on Leisure
Panel A: Women's Time Spent on Leisure		
Spouse's pension eligibility	0.038 (0.05)	0.085 (0.32)
Controls	Yes	Yes
Year $\times$ Province	Yes	Yes
Observations	2,173	2,172
Ineligible Group Mean	0.274	1.518
Adj R-squared	0.046	0.044
Panel B: Men's Time Spent on Leisure		
Own pension eligibility	-0.013 (0.05)	-0.015 (0.30)
Controls	Yes	Yes
Year $\times$ Province	Yes	Yes
Observations	2,111	2,111
Ineligible Group Mean	0.217	1.134
Adj R-squared	0.016	0.016

The dependent variable is whether the individual has leisure activities and how many days per week spent on leisure activities. Variable of interest is husband's pension eligibility. Standard errors are reported in parentheses and clustered at the individual level. * $p < .1$, ** $p < .05$, *** $p < .01$.

Table 15: Single regression discontinuity design on couples' leisure activities: husbands never non-agricultural pre-60

	Have Leisure Activities	Day Spent on Leisure
Panel A: Women's Time Spent on Leisure		
Spouse's pension eligibility	0.158** (0.07)	1.147** (0.45)
Controls	Yes	Yes
Year $\times$ Province	Yes	Yes
Observations	1,246	1,245
Ineligible Group Mean	0.254	1.342
Adj R-squared	0.046	0.062
Panel B: Men's Time Spent on Leisure		
Own pension eligibility	0.127* (0.08)	0.536 (0.48)
Controls	Yes	Yes
Year $\times$ Province	Yes	Yes
Observations	1,263	1,263
Ineligible Group Mean	0.258	1.440
Adj R-squared	0.020	0.022

The dependent variable is whether the individual has leisure activities and how many days per week spent on leisure activities. Variable of interest is husband's pension eligibility. Standard errors are reported in parentheses and clustered at the individual level. * $p < .1$, ** $p < .05$, *** $p < .01$.

References

- Atalay, K., G. F. Barrett, and P. Siminski (2019). Pension incentives and the joint retirement of couples: Evidence from two natural experiments. *Journal of Population Economics* 32, 735–767.
- Barrientos, A., M. Gorman, and A. Heslop (2003, March). Old Age Poverty in Developing Countries: Contributions and Dependence in Later Life. *World Development* 31(3), 555–570.
- Battistin, E., A. Brugiavini, E. Rettore, and G. Weber (2009). The retirement consumption puzzle: evidence from a regression discontinuity approach. *American Economic Review* 99(5), 2209–2226.
- Becker, G. S. (1965). A theory of the allocation of time. *The Economic Journal* 75(299), 493–517.
- Blau, D. M. (1998). Labor force dynamics of older married couples. *Journal of Labor Economics* 16(3), 595–629.
- Bonsang, E. and A. Van Soest (2020). Time devoted to home production and retirement in couples: a panel data analysis. *Labour Economics* 65, 101810.
- Bredtmann, J., S. Otten, and C. Rulff (2018). Husband's unemployment and wife's labor supply: The added worker effect across europe. *ILR Review* 71(5), 1201–1231.

- Burbidge, J. B., L. Magee, and A. L. Robb (1988). Alternative transformations to handle extreme values of the dependent variable. *Journal of the American statistical Association* 83(401), 123–127.
- Cai, F., J. Giles, P. O’Keefe, and D. Wang (2012). Old-age support in rural china: Challenges and prospects. *Washington, DC: World Bank*.
- Chan, K. W. (2009). The chinese hukou system at 50. *Eurasian Geography and Economics* 50(2), 197–221.
- Chan, K. W. and L. Zhang (1999). The hukou system and rural-urban migration in china: Processes and changes. *The China Quarterly* 160, 818–855.
- Chen, J. and J. Roth (2023, December). Logs with zeros? some problems and solutions*. *The Quarterly Journal of Economics* 139(2), 891936.
- Chen, Y., L. Zhao, Y. Fan, and B. Xie (2021). Does the new rural pension scheme improve residents’ livelihoods? empirical evidence from northwestern china. *PLoS ONE* 16(4), e0250130.
- Cheng, T. and M. Selden (1994). The Origins and Social Consequences of China’s Hukou System. *The China Quarterly* 139, 644–668.
- Chiappori, P.-A. (1992). Collective labor supply and welfare. *Journal of Political Economy* 100(3), 437–467.
- Ciani, E. (2016). Retirement, pension eligibility and home production. *Labour Economics* 38, 106–120.
- Deng, Y., H. Fang, K. Hanewald, and S. Wu (2023). Delay the pension age or adjust the pension benefit? implications for labor supply and individual welfare in china. *Journal of Economic Behavior & Organization* 212, 11921215.
- Fang, H. and J. Feng (2018, September). The chinese pension system. Working Paper 25088, National Bureau of Economic Research.
- García, A. N. and A. van Soest (2022). Joint retirement behaviour and pension reform in the netherlands. *The Journal of the Economics of Ageing* 23, 100401.
- García-Miralles, E. and J. M. Leganza (2024). Joint retirement of couples: Evidence from discontinuities in Denmark. *Journal of Public Economics* 230, 105036.
- Giles, J., X. Lei, G. Wang, Y. Wang, and Y. Zhao (2023). One country, two systems: evidence on retirement patterns in china. *Journal of pension economics & finance* 22(2), 188–210.
- Giles, J., D. Wang, and W. Cai (2011). The labor supply and retirement behavior of china’s older workers and elderly in comparative perspective. Technical Report 5853, World Bank.

- Gong, X. (2011). The added worker effect for married women in australia. *Economic Record* 87(278), 414–426.
- Gronau, R. (1973). The intrafamily allocation of time: The value of the housewives' time. *American Economic Review* 63(4), 634–651.
- Gronau, R. (1977). Leisure, home production, and work—the theory of the allocation of time revisited. *Journal of Political Economy* 85(6), 1099–1123.
- Gustman, A. L. and T. Steinmeier (2009). Integrating retirement models. Technical report, National Bureau of Economic Research.
- Gustman, A. L. and T. L. Steinmeier (2000). Retirement in dual-career families: a structural model. *Journal of Labor Economics* 18(3), 503–545.
- Hahn, J. and G. Kuersteiner (2002). Asymptotically unbiased inference for a dynamic panel model with fixed effects when both n and t are large. *Econometrica* 70(4), 1639–1657.
- Humphrey, D. D. (1940). Alleged "additional workers" in the measurement of unemployment. *Journal of Political Economy* 48(3), 412–419.
- Hurd, M. D. (1988, December). The Joint Retirement Decision of Husbands and Wives. NBER Working Papers 2803, National Bureau of Economic Research, Inc.
- Imbens, G. and K. Kalyanaraman (2012). Optimal bandwidth choice for the regression discontinuity estimator. *The Review of Economic Studies* 79(3), 933–959.
- Imbens, G. and S. Wager (2019). Optimized regression discontinuity designs. *Review of Economics and Statistics* 101(2), 264–278.
- Imbens, G. W. and T. Lemieux (2008). Regression discontinuity designs: A guide to practice. *Journal of econometrics* 142(2), 615–635.
- Jacobson, L. S., R. J. LaLonde, and D. G. Sullivan (1993). Earnings losses of displaced workers. *The American economic review*, 685–709.
- Johnsen, J. V., K. Vaage, and A. Willén (2022). Interactions in public policies: Spousal responses and program spillovers of welfare reforms. *The Economic Journal* 132(642), 834–864.
- Kruse, H. (2021). Joint retirement in couples: Evidence of complementarity in leisure. *The Scandinavian journal of economics* 123(3), 995–1024.
- Lalive, R. and P. Parrotta (2017). How does pension eligibility affect labor supply in couples? *Labour Economics* 46, 177–188.
- Layard, R. and S. J. Nickell (1986). Unemployment in Britain. *Economica* 53(210), S121–S169.

- Lee, D. S. and T. Lemieux (2010, June). Regression discontinuity designs in economics. *Journal of Economic Literature* 48(2), 281355.
- Lei, X., J. Giles, Y. Hu, A. Park, J. Strauss, and Y. Zhao (2012). Patterns and correlates of intergenerational nontime transfers: Evidence from charls. In N. R. Council (Ed.), *Aging in Asia: Findings from New and Emerging Data Initiatives*, pp. 207–228. Washington, DC: The National Academies Press.
- Long, C. D. (1958). *The Labor Force under Changing Income and Employment*. Princeton University Press.
- Lundberg, S. (1985). The added worker effect. *Journal of Labor Economics* 3(1), 11–37.
- Michaud, P.-C., A. Van Soest, and L. Bissonnette (2020). Understanding joint retirement. *Journal of economic behavior & organization* 173, 386–401.
- Michaud, P.-C. and F. Vermeulen (2011). A collective labor supply model with complementarities in leisure: Identification and estimation by means of panel data. *Labour Economics* 18(2), 159–167.
- Mincer, J. (1962). Labor force participation of married women: A study of labor supply. In H. G. Lewis (Ed.), *Aspects of Labor Economics*, pp. 63–105. Princeton, NJ: Princeton University Press.
- Moghadam, H. M., P. A. Puhani, and J. Tyrowicz (2024). Pension reforms and couples labour supply decisions. *Labour Economics* 91, 102627.
- Ning, M., J. Gong, X. Zheng, and J. Zhuang (2016). Does new rural pension scheme decrease elderly labor supply? evidence from charls. *China Economic Review* 41, 315–330.
- Rogerson, R. and J. Wallenius (2019). Household time use among older couples: evidence and implications for labor supply parameters. *The Quarterly Journal of Economics* 134(2), 1079–1120.
- Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of educational Psychology* 66(5), 688.
- Seibold, A. (2021). Reference points for retirement behavior: Evidence from german pension discontinuities. *American Economic Review* 111(4), 1126–1165.
- Stancanelli, E. and A. V. Soest (2012). Retirement and home production: A regression discontinuity approach. *American Economic Review* 102(3), 600–605.
- Stephens Jr, M. (2002). Worker displacement and the added worker effect. *Journal of Labor Economics* 20(3), 504–537.
- Wang, D. (2006, February). China’s urban and rural old age security system: Challenges and options. *China & World Economy* 14(1), 102–116.

- Wang, F.-L. (2005). *Organizing through Division and Exclusion: China's Hukou System*. Stanford University Press.
- Woytinsky, W. S. (1942). *Three Aspects of Labor Dynamics*. Washington, DC: Social Science Research Council.
- Wu, X. and D. J. Treiman (2004). The household registration system and social stratification in china: 1955-1996. *Demography* 41(2), 363–384.
- Zhao, Y., Y. Hu, J. P. Smith, J. Strauss, and G. Yang (2014). Cohort Profile: The China Health and Retirement Longitudinal Study (CHARLS). *International Journal of Epidemiology* 43(1), 61–68.